Immunosuppressors & Immunostimulators

Broad-Spectrum Immunosuppressants

Corticosteroids

Maintenance Transplant Backbone

Calcineurin Inhibitors
- Cyclosporine
- Tacrolimus

mTOR Inhibitors
- Sirolimus
- Everolimus

Cellular Antiproliferatives

Azathioprine
Mycophenolate (MPA)
Leflunomide / Teriflunomide

Targeted Biologics & Small Molecules

Basiliximab
rATG / Alemtuzumab
Belatacept / Abatacept
Rituximab
Bortezomib
JAK inhibitors
Anti-cytokines
IVIG

Immunostimulants

Vaccines
Filgrastim / GM-CSF
Interferons
Imiquimod

Oncology Immunotherapy

Checkpoint Inhibitors
CAR-T Cell Therapy

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Pharmacological Taxonomy

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"Immunomodulators are tools for resetting an imbalanced immune system by targeting specific points in the immune network for each disease, while accepting trade-offs between efficacy and the risks of infection, cancer, and site-specific toxicities."

Immunosuppressors

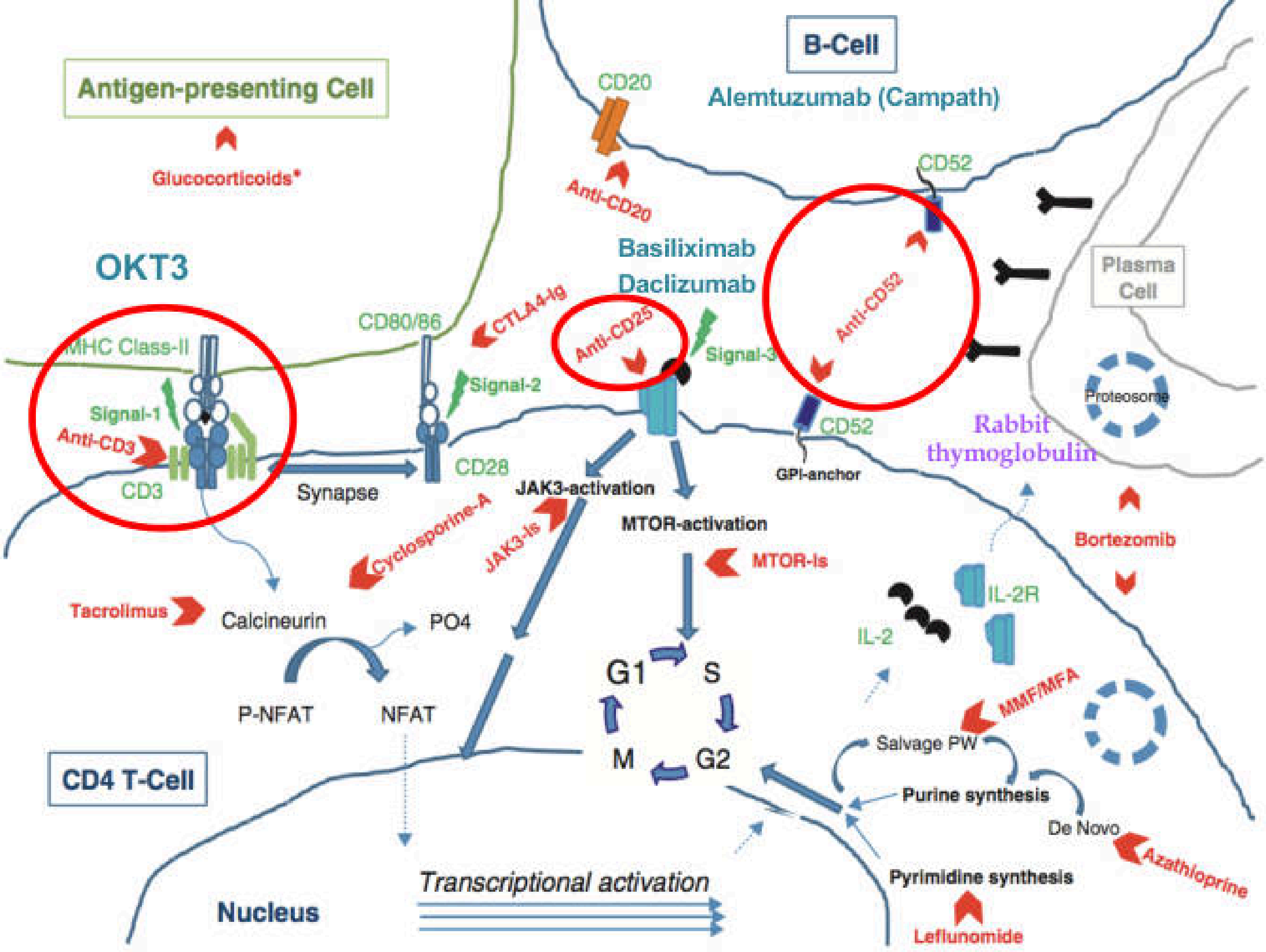
- are substances that reduce or inhibit the immune response, often used in organ transplantations, autoimmune diseases, and conditions where the immune system is overactive
  - Lymphocyte Depletion Agents
  - Calcineurin Inhibitors
  - Corticosteroids
  - Antimetabolites
  - mTOR Inhibitors
  - Biologics
  - Janus Kinase Inhibitors (JAK inhibitors)
  - Antibodies Against Cytokines

Immunosuppressors

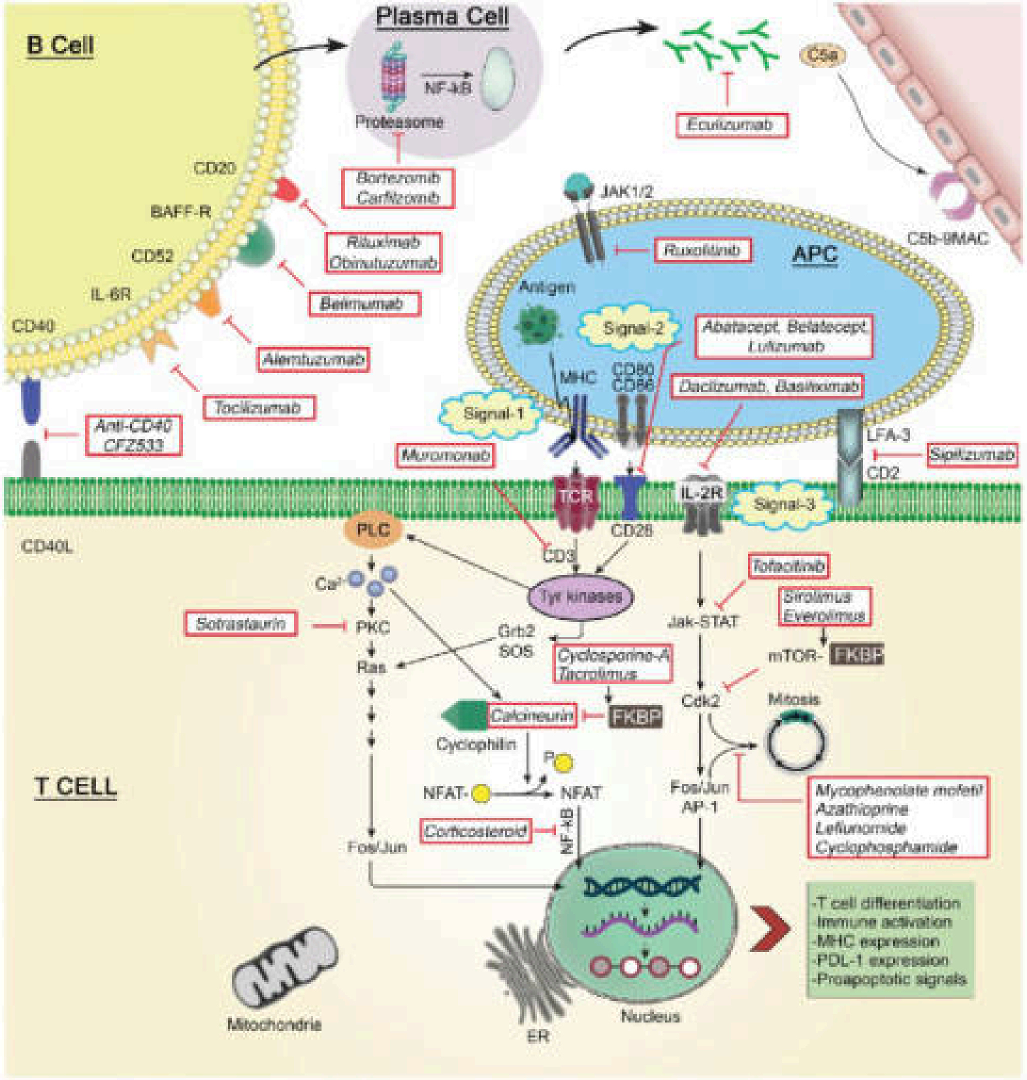
Immunosuppressant drugs suppress or reduce the strength of the immune system. Some types help prevent organ transplant rejection, while others treat autoimmune disorders like lupus or psoriasis

Intracellular (initiation)	Antimetabolites			*purine synthesis inhibitors:* Azathioprine · Mycophenolic acid *pyrimidine synthesis inhibitors:* Leflunomide · Teriflunomide *antifolate:* Methotrexate
	Macrolides/ other IL-2 inhibitors			*FKBP/Cyclophilin/Calcineurin:* Tacrolimus · Ciclosporin · Pimecrolimus Abetimus · Gusperimus
	IMiDs			Lenalidomide · Pomalidomide · Thalidomide
Intracellular (reception)	IL-1 receptor antagonists			Anakinra
	mTOR			Sirolimus · Everolimus · Ridaforolimus · Temsirolimus · Umirolimus · Zotarolimus
Extracellular	Antibodies	Monoclonal	Serum target (noncellular)	*Complement component 5* (Eculizumab) · *TNF* (Adalimumab · Afelimomab · Certolizumab pegol · Golimumab · Infliximab · Nerelimomab) · *Interleukin 5* (Mepolizumab) · *Immunoglobulin E* (Omalizumab) *Interferon* (Faralimomab) *IL-6* (Elsilimomab) *IL-12* and *IL-23* (Lebrikizumab · Ustekinumab)
			Cellular target	*CD3* (Muromonab-CD3 · Otelixizumab · Teplizumab · Visilizumab) · *CD4* (Clenoliximab · Keliximab · Zanolimumab) · *CD11a* (Efalizumab) · *CD18* (Erlizumab) · *CD20* (Obinutuzumab · Rituximab · Ocrelizumab · Pascolizumab) · *CD23* (Gomiliximab · Lumiliximab) · *CD40* (Teneliximab · Toralizumab) · *CD62L/L-selectin* (Aselizumab) · *CD80* (Galiximab) · *CD147/Basigin* (Gavilimomab) · *CD154* (Ruplizumab) *BLyS* (Belimumab · Blisibimod) · *CTLA-4* (Ipilimumab · Tremelimumab) · *CAT* (Bertilimumab · Lerdelimumab · Metelimumab) · *Integrin* (Natalizumab) · *Interleukin-6 receptor* (Tocilizumab) · *LFA-1* (Odulimomab) *IL-2 receptor/CD25* (Basiliximab · Daclizumab · Inolimomab) *T-lymphocyte* (Zolimomab aritox)
			Unsorted	Atorolimumab · Cedelizumab · Fontolizumab · Maslimomab · Morolimumab · Pexelizumab · Reslizumab · Rovelizumab · Siplizumab · Talizumab · Telimomab aritox · Vapaliximab · Vepalimomab
		Polyclonal		Anti-thymocyte globulin · Anti-lymphocyte globulin
	-cept (Fusion)			*CTLA-4* (Abatacept · Belatacept) · *TNF inhibitor* (Etanercept · Pegsunercept) · Aflibercept · Alefacept · Rilonacept

https://en.wikipedia.org/wiki/ATC_code_L04

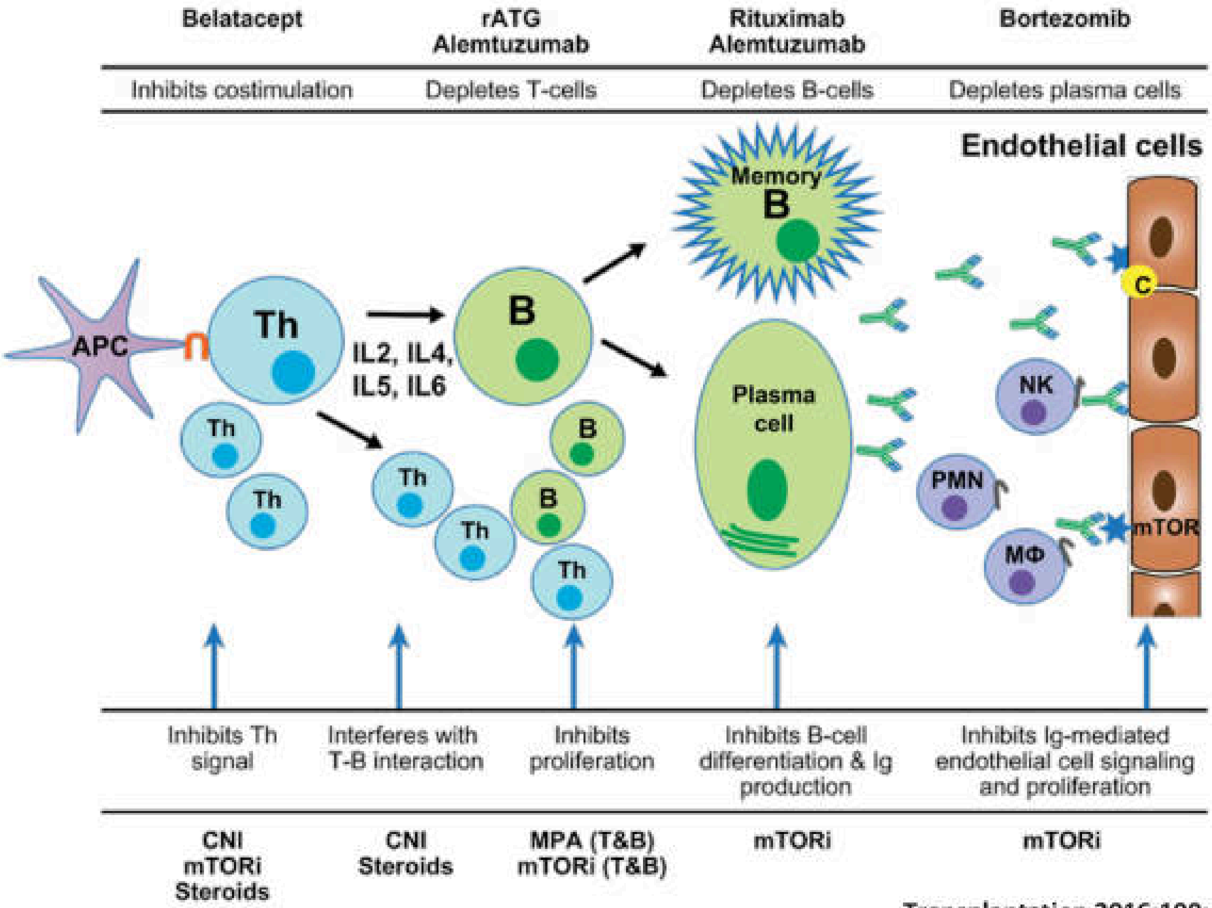


Antigen-presenting Cell
Glucocorticoids*
B-Cell
CD20
Alemtuzumab (Campath)
Anti-CD20
CD52
OKT3
Basiliximab
Daclizumab
Plasma Cell
CTLA4-Ig
CD80/86
Anti-CD25
Anti-CD52
MHC Class-II
Signal-3
Signal-2
Signal-1
Proteosome
Anti-CD3
CD52
Rabbit thymoglobulin
CD28
CD3
Synapse
JAK3-activation
GPI-anchor
MTOR-activation
Cyclosporine-A
JAK3-Is
Bortezomib
MTOR-Is
Tacrolimus
Calcineurin
PO4
IL-2R
IL-2
G1
S
MMF/MFA
P-NFAT
NFAT
M
G2
Salvage PW
CD4 T-Cell
Purine synthesis
De Novo
Azathioprine
Pyrimidine synthesis
Transcriptional activation
Nucleus
Leflunomide



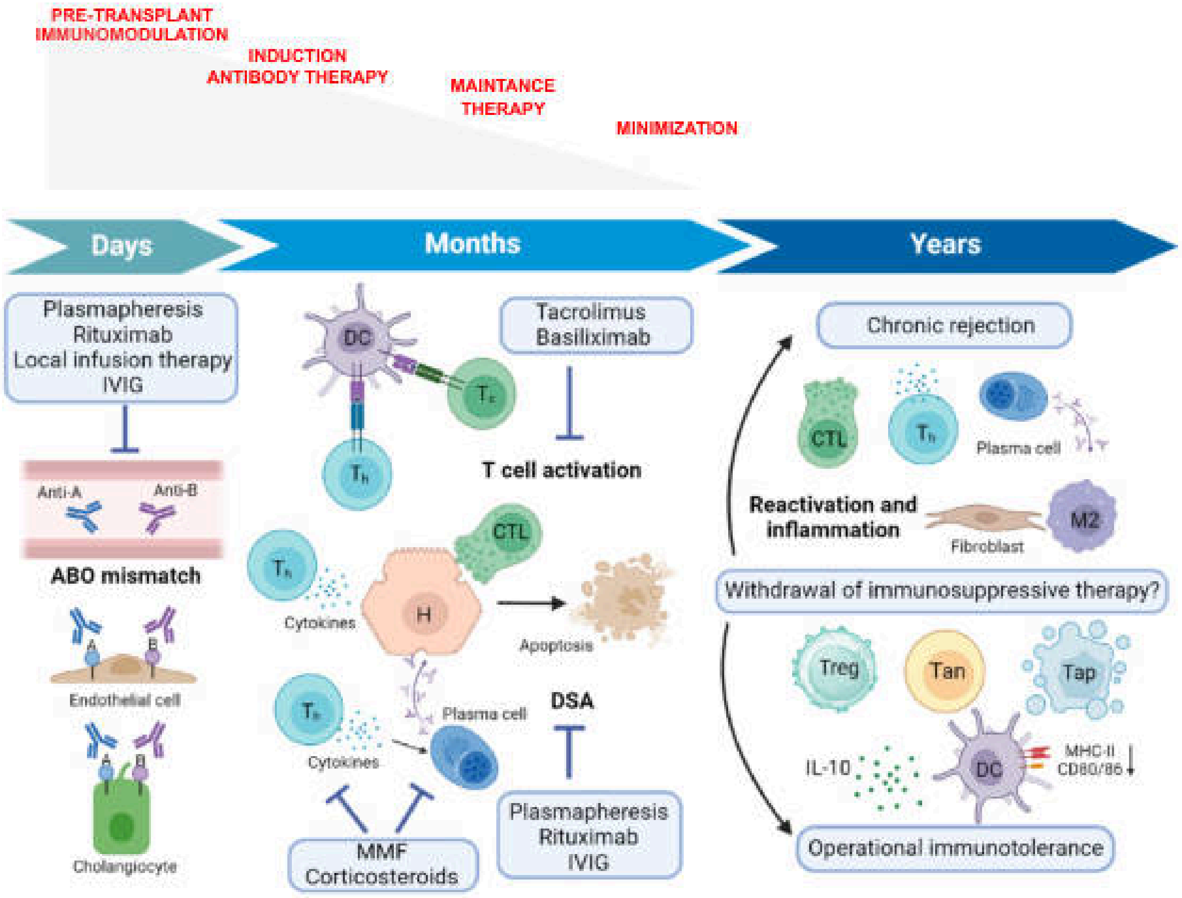
B Cell
Plasma Cell
NF-kB
Proteasome
Bortezomib
Carfilzomib
Eculizumab
C5a
C5b-9MAC
CD20
BAFF-R
CD52
IL-6R
CD40
Rituximab
Obinutuzumab
Belimumab
Alemtuzumab
Tocilizumab
Anti-CD40
CFZ533
JAK1/2
Ruxolitinib
APC
Antigen
Signal-2
Abatacept, Belatacept,
Lulizumab
MHC
CD80
CD86
Daclizumab, Basiliximab
Signal-1
LFA-3
Siplizumab
CD2
Muromonab
TCR
IL-2R
Signal-3
CD28
CD3
PLC
CD40L
Ca
Tyr kinases
Tofacitinib
Sirolimus
Everolimus
Jak-STAT
Sotrastaurin
PKC
Grb2
SOS
Ras
Cyclosporine-A
Tacrolimus
mTOR-
FKBP
Cdk2
Mitosis
Calcineurin
FKBP
Cyclophilin
P
NFAT-
NFAT
Fos/Jun
AP-1
Mycophenolate mofetil
Azathioprine
Leflunomide
Cyclophosphamide
T CELL
Corticosteroid
NF-kB
Fos/Jun
-T cell differentiation
-Immune activation
-MHC expression
-PDL-1 expression
-Proapoptotic signals
Mitochondria
ER
Nucleus

Mode of action of key immunosuppressant



Transplantation 2016;100: 39–53

Stages of Immunosuppressive Therapy



World J Transplant 2024 March 18; 14(1): 89772

Drug class	Drugs	Site of action/mechanism
Agents used in induction regimens		
Lymphocyte depleting antibodies	OKT3 (monoclonal)	Signal-1 → anti-CD3 T-cell depletion/lysis
	Rabbit thymoglobulin	Multiple surface receptors — T-cell depletion/lysis
	Campath	Anti-CD52 — T-cell, B-,cell NK-cell lysis
Lymphocyte non-depleting Ab	Daclizumab	Signal-3 → anti-CD25 (α-chain of IL-2 receptor) competes for IL-2
	Basiliximab	Signal-3 → anti-CD25 (α-chain of IL-2 receptor) competes for IL-2
Agents used for maintenance immunosuppression		
Co-stimulation blockade	Abatacept	Signal-2 → competitive inhibition of costimulation → anergic signal
	Belatacept	Signal-2 → competitive inhibition of costimulation → anergic signal
Calcineurin inhibitors	Cyclosporine-A	Signal-1 → binds cyclophilin and inhibits calcineurin mediated de-phosphorylation of NFAT and transcription
	Tacrolimus	Signal-1 → as cyclosporine but binds FK-binding protein-2 to inhibit calcineurin
MTOR-inhibitors	Sirolimus	Signal-2 → inhibits cellular MTOR-activation and Cell proliferation
	Everolimus	Signal-2 → as above
Antiproliferative agents	Mycophenolate mofetil	IMP-dehydrogenase-II inhibitor → inhibits purine nucleoside synthesis through Salvage PW
	Mycophenolic acid (enteric coated)	IMPDH → as above
	Azathioprine	*Multiple effects*: amido-phosphoribosyltransferase, IMPDH, adenylosuccinate synthase, RNA inhibition, pro-apoptosis
	Leflunomide	Mitochondrial dihydroorotate dehydrogenase inhibitor → pyrimidine nucleotide synthesis and T-cell proliferation
Glucocorticoids	Prednisone	*Multiple effects*: inhibits innate immune responses, antigen presentation, TH-1 differentiation, T-cell proliferation
Agents used in desensitization protocols		
Intravenous gamma globulin (IVIG)		*Multiple putative mechanisms*: idiotype-antiidiotype complex clearance, diversion of complement from allograft, immunoregulatory effects of sialylated IgG, increased Fc-rRIIB and macrophage inhibition
Rituximab		Anti-CD20 antibody. Antibody-dependent complement-mediated B-cell lysis. B-cell apoptosis
Bortezomib		26S proteosome inhibition. T-cell, B-cell and Plasma cell affected. NFKB depletion, Endoplasmic reticulum stress and apoptosis

Categories of Agents

Induction Immunosuppression

- High levels of immunosuppression in the peri- and immediate- post-transplantation period
- Goal
  - Reduce risk of acute rejection
  - Reduce memory cells primed to respond to donor antigen?
- Two classes of agents
  - lymphocyte depletionary: OKT3, antithymocyte globulin (ATG), alemtuzumab
  - Non-depletionary: basiliximab, daclizumab

Magee and Pascual, Arch Intern Med, July 2004

Rabbit Antithymocyte Globulin (ATG, Thymoglobulin®)

- Dosing: 1.5 mg/kg IV daily (rounded to nearest 25 mg vial) – induction 5 mg/kg total; rejection 4-7 doses
- Dose adjustment daily based on WBC/Platelet
- Mechanism of Action:
  - Antibodies to CD2, CD3, CD4, CD8, CD11a, CD18, CD25, HLA-DR, bind to cell surface receptors and cause opsonization
  - Made from immunization of rabbits with human thymocytes
- Infusion-related reactions reduced by pre-medication

Halloran, NEJM, December 2004

Alemtuzumab (Campath 1H®)

- MOA: Humanized antibody to CD52 (5% of all lymphocyte surface antigens) initiating cell lysis
  - T lymphocytes (about 500K molecules per cell)
  - B lymphocytes
  - NK cells
  - Monocytes, macrophages, eosinophils
- Dosing: 30 mg prior to surgery
- Side effects: infusion-related reactions, neutropenia, thrombocytopenia, autoimmunity

Bartosh, Am J Transplantation, 2005, Morris and Russell, Transplantation, May 2006, Halloran, NEJM, December 2004

Basiliximab (Simulect®)

- MOA:
  - Basiliximab chimeric monoclonal antibody
  - Competitively inhibits α chain of IL-2 receptor on activated T-cells
- Place in therapy:
  - Equal graft and patient survival rates as with ATG
  - Less ability to allow steroid free / Calcineurin Inhibitor avoidance protocols as depletional agents

Chapman and Keating, Drugs, 2003

Which induction agent ?

- rATG – deplete T cells >> B cells
- Alemtuzumab – deplete T cells = B cells = APC
- Basiliximab – non deplete lymphocyte
- Alem. and rATG – choice in high risk patients
- Basiliximab – low risk recipient eg., older, history of HCV, HBV and HIV
- Lowest risk of graft failure
  - Alemtuzumab << rATG << Basiliximab

Current Issues and Future Direction in Kidney Transplantation
http://dx.doi.org/10.5772/54865

Antibody Mediated Rejection (AMR) Treatment Options

- B-Cell targeted therapies
  - Low dose IVIG + Plasmapheresis ± Rituximab
  - Bortezomib (Velcade)

UW Guidelines, 2009

Intravenous immunoglobulin: IVIG

- MOA:
  - Feedback inhibition of plasma cells
  - When combined with plasmapheresis prevents plasma cell overproduction
- Dosing:
  - High dose is 2 gm/kg (max 140 gm)
  - Low dose is 100 mg/kg (max of 7 gm)
- Adverse Effects:

  - Osmotic nephropathy, headache, thrombotic complications, aseptic meningitis, bronchospasm, hemolytic anemia

Jordan, AJT, 2011

Selection of Product for individual patients

Table 1. Clinical Considerations: Matching the IVIG With the Patient Profile

Patient Risk Factors	IVIG Characteristics					
	Volume Load	Sugar Content	Sodium Content	Osmolality	pH	IgA
Cardiac impairment	X		X	X		
Renal dysfunction	X	X	X	X		
Anti-IgA antibodies						X
Thromboembolic risk	X		X	X		
Prediabetes		X				
Geriatric patients	X	X	X	X		
Neonates/pediatrics	X		X	X	X	

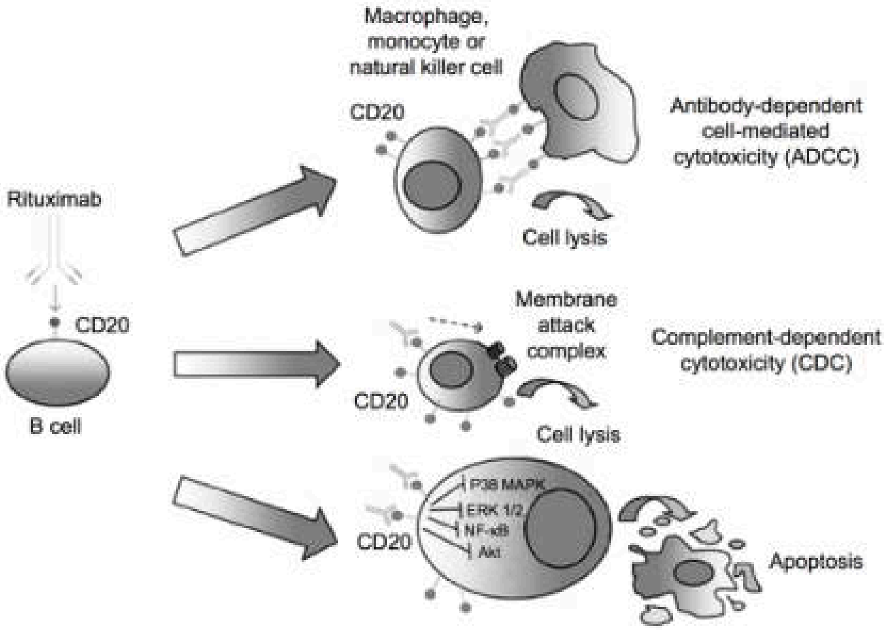
IgA, immunoglobulin A

Infusion time: IVIG 10% faster than IVIG 5%

Rituximab

- MOA:

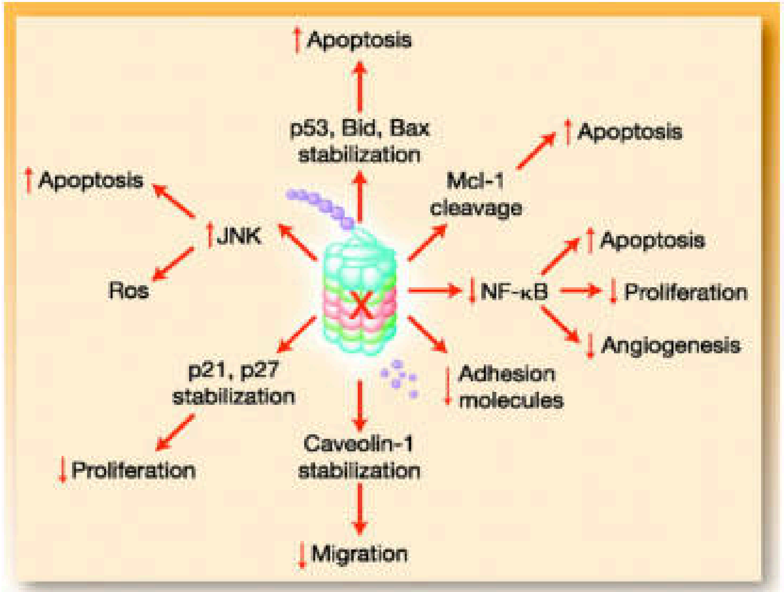
  Chimeric monoclonal anti-CD20 antibody present on mature B cells; question on plasma cells
- Dosing: 375 mg/m^2 (max 750 mg) IV infusion
- Adverse Effects: infusion-related reactions, thrombocytopenia



UW Guidelines, 2009

Bortezomib

- MOA:
  - Bortezomib reversibly binds to the chymotrypsin-like subunit of the 26S proteasome, resulting in its inhibition and preventing the degradation of various pro-apoptotic factors.
  - The accumulation will eventually activate the programmed cell death via caspase-mediated pathways in the neoplastic cells that are usually dependent on the suppression of pro-apoptotic pathways for their proliferation and survival.



- Dosing:
  - 1.4 mg/kg x 4 doses
- Adverse Effects:

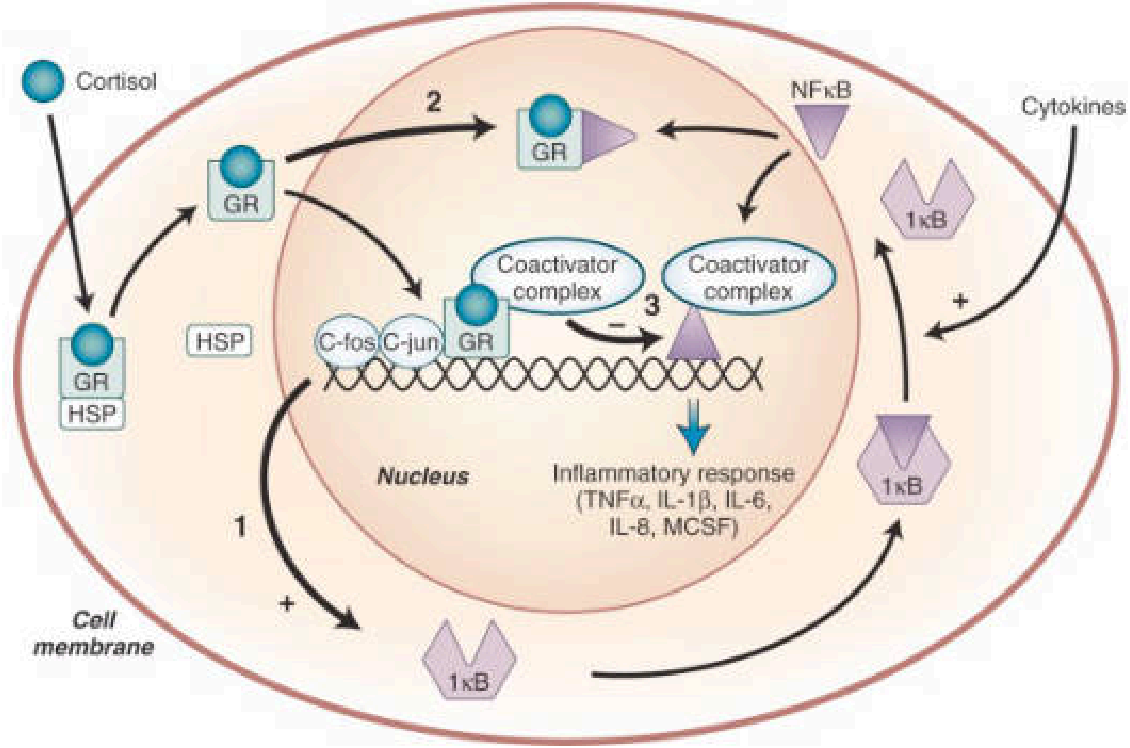
  - Hypotension, peripheral neuropathy, GI (N/V/D), thrombocytopenia, neutropenia

UW Guidelines, 2009

Maintenance Immunosuppression

- The goals of maintenance immunosuppression are to prevent ACR episodes and optimize long-term patient and allograft survival rates.
- Used begins either simultaneously with or immediately after the initiation of induction therapy.
- Maintenance regimens generally combine a primary immunosuppressant with 1 or 2 adjunctive agents.

Corticosteroids: MOA



Corticosteroids affect nearly every facet of immune function, although less inhibition of humoral arm than cell-mediated arm; they also induce apoptosis in rapidly-dividing leukocytes

Steroids

- MOA:
  - Both humoral and cellular immunity is disrupted
  - Inhibits cytokine (IL-1,IL-2, IL-6, TNF, IFN-gamma) gene transcription
- Adverse Effects:
  - Acute: Multiple - GI, insomnia, leukocytosis, mood changes
  - Chronic: new onset diabetes (NODAT), weight gain, osteoporosis, hypertension, hyperlipidemia, cushingoid features, acne, hirsutism, impaired wound healing, cataracts, peptic ulcer disease (PUD), myopathy

Magee and Pascual, Arch Intern Med, July 2004
Halloran, NEJM, December 2004

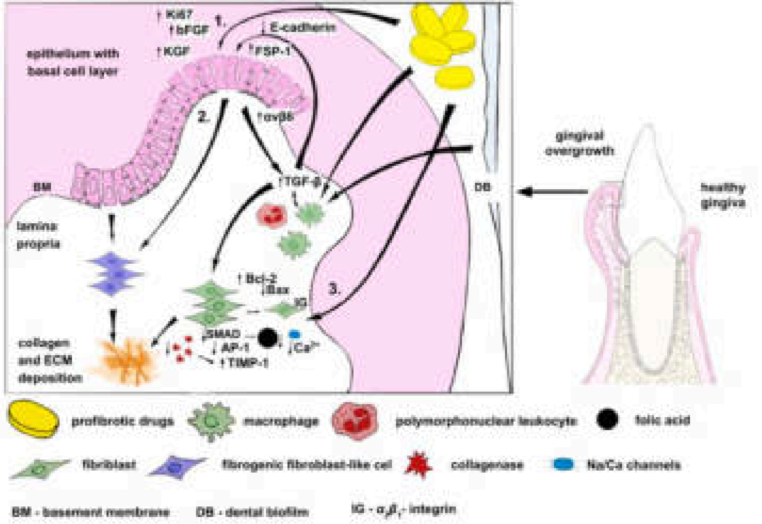
Steroids

- Place in therapy:
  - Use during induction phase to reduce inflammation within graft
  - Prednisone withdrawal > 3 months increases risk of rejection (9.8% vs 30.8% risk of AR at 1 year)
  - Use when unable to provide other immunosuppressants
- Dosing – varies widely by center / induction:
  - Adult –
    - Dexamethasone 100 mg IV on POD #0
    - Dexamethazone 50 mg IV on POD #1
    - Prednisone 90 mg PO POD #2
    - Prednisone 60 mg on PO POD #3
    - Prednisone 30 mg starting on PO POD #4 until seen in clinic

Ahsan, Transplantation, December 1999

Calcineurin Inhibitor: Cyclosporine

- MOA:
  - Binds to cyclophilin, which binds to calcineurin, which inhibits dephosphorylation and transport of nuclear factor of activated T cells, resulting in decreased production of IL-2 and other cytokines
- Adverse effects:
  - Nephrotoxicity, hypertension, hyperkalemia, neurotoxicity (encephalopathy, tremor), hyperlipidemia, gout, hirsutism, gingival hyperplasia, hypomagnesemia, hypophosphatemia, NODAT, osteopenia



Magee and Pascual, Arch Intern Med, July 2004; Halloran, NEJM, December 2004, Int. J. Mol. Sci. 2023, 24, 5448

Calcineurin Inhibitor: Tacrolimus

- MOA:
  - Binds to FK binding proteins, which bind to calcineurin, which inhibits dephosphorylation and transport of nuclear factor of activated T cells (NFAT), resulting in decreased production of IL-2 and other cytokines
- More potent than cyclosporine (20-30X)
- Drug interactions
  - Metabolized by liver by cytochrome P450 system
  - Avoid grapefruit / pomegranate juice
- Adverse Effects:
  - Nephrotoxicity, NODAT, HTN, hyperkalemia, neurotoxicity (encephalopathy, tremors), hyperlipidemia, alopecia, hypomagnesemia, hypophosphatemia, diarrhea, osteopenia

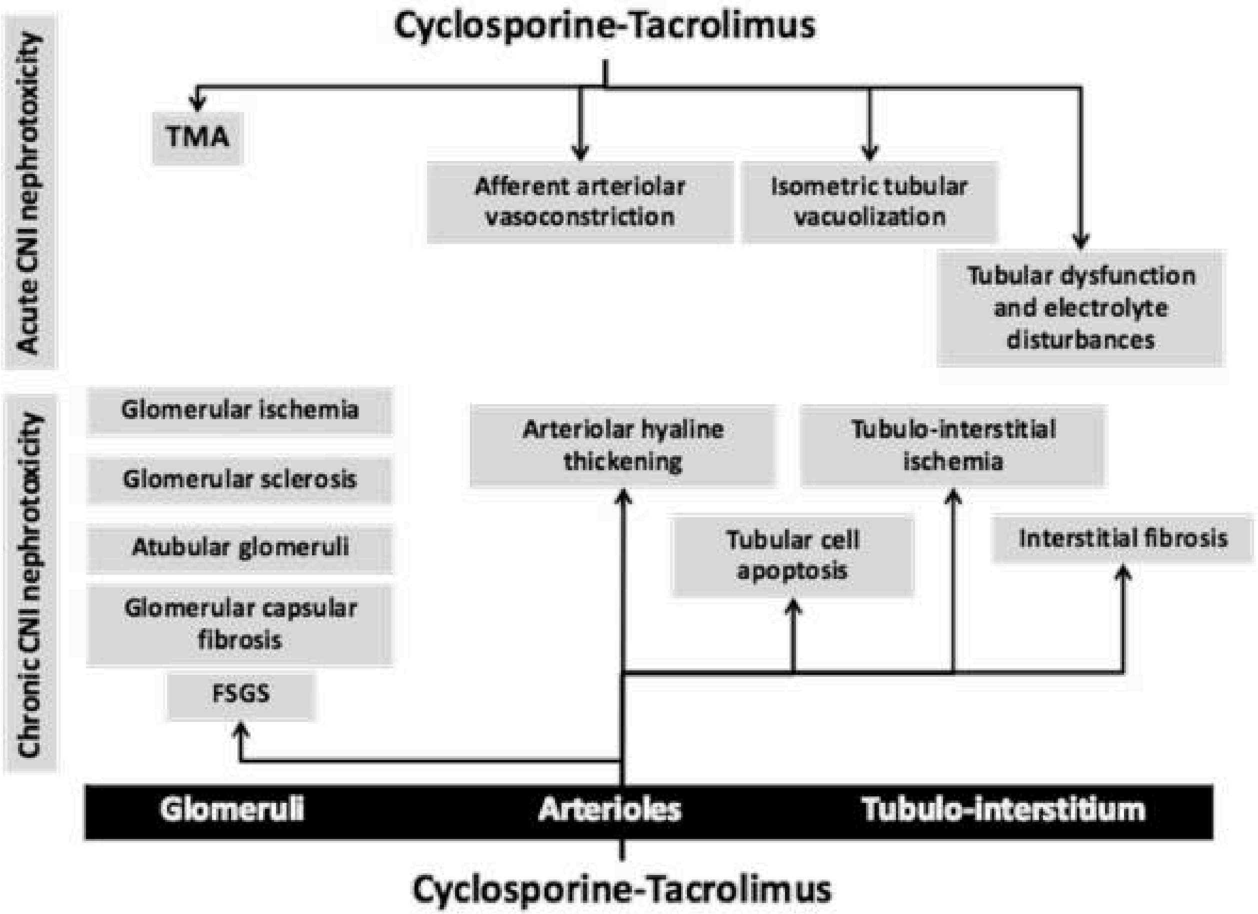
Magee and Pascual, Arch Intern Med, July 2004; Halloran, NEJM, December 2004

Calcineurin Inhibitor: Tacrolimus

- Administration
  - Ideal is to take on an empty stomach, weigh against compliance
  - Cannot be with calcium, magnesium, iron, or antacids
  - Trough levels will vary depending on the time frame from transplant

- Dosing
  - Adult 0.1 mg/kg/dose split BID
  - May need higher doses in African American

Calcineurin Inhibitor: Nephrotoxicity



Antiproliferative Agent: Azathioprine

- MOA:
  - Inhibit B and T lymphocyte proliferation by antagonizing purine metabolism
- No longer commonly used
- Important drug interaction with **allopurinol**
- Adverse Effects:
  - Bone marrow suppression, malignancy, rash, N/V/D, pancreatitis

Magee and Pascual, Arch Intern Med, July 2004; Halloran, NEJM, December 2004

Antiproliferative Agent: Mycophenolate

- MOA:
  - Non-competitive, selective, reversible inosine monophosphate dehydrogenase inhibitor (IMPDH)
  - Rate limiting step for guanine nucleotides in T-cells and B-cells (dependent on *de novo* pathway)
- Adverse Effects:
  - GI (N/V/Diarrhea), bone marrow suppression, CMV, GI bleed, lymphomas

- 720 mg BID (2 x 360 mg)
- 180 mg and 360 mg tablets
- Enteric coated
  - Delays release until it reaches the small intestine
    - Mycophenolate sodium

- 1000 mg BID (2 x 500 mg)
- 250 mg cap / 500 mg tablets
- Immediate release
  - Releases in stomach
    - Mycophenolate mofetil

Magee and Pascual, Arch Intern Med, July 2004
Halloran, NEJM, December 2004

Antiproliferative Agent: Mycophenolate

- Pharmacokinetics –
  - Rapidly and completely absorbed
  - Enterohepatic recycling
- Monitoring -
  – APOMYGRE study showed benefit
  – Rarely use trough levels
- Administration –
  – Better tolerated with food
  – Cannot be taken with calcium, magnesium, iron, or antacids

m-TOR Inhibitors: Sirolimus / Everolimus

- MOA:
  - Bind to FK binding protein, which inhibits target of rapamycin (TOR) - blocks intracellular signals past IL-2 receptor
- Adverse Effects:
  - NODAT, thrombocytopenia, anemia, leukopenia, interstitial pneumonitis, hypokalemia, delayed wound healing (lymphocele), proteinuria, TMA, hyperlipidemia
  - Delays Acute Tubular Necrosis (ATN) recovery

Magee and Pascual, Arch Intern Med, July 2004
Halloran, NEJM, December 2004

m-TOR Inhibitor: Sirolimus

- Place in therapy:
  - Reduced use as initial maintenance immunosuppression
  - Sirolimus had higher rates of rejection and higher rates of treatment failure
  - Lower rates of malignancies
  - Everolimus, higher rates of rejection and discontinuation due to adverse effects

Co-stimulatory Blockade: Belatacept

- MOA:
  - Fusion protein that acts as a potent antagonist of B7 ligands (B7-1 [CD80] and B7-2 [CD86]) on APC's which activates CD28 in naïve T-cells
  - Does not eliminate or inhibit memory T-cells
  - Blocks "Signal 2" in T-cell activation prevent IL-2 release
- Adverse Effects:
  - Posttransplant lymphoproliferative disorder; PTLD (with CNS involvement), infusion reactions, aphthous stomatitis, proteinuria
- Dosing:
  - First intermittent infusion maintenance therapy for renal transplant patients
  - May have a role in noncompliant patients or may allow for CNI and steroid free protocols

Typical Transplant Regimens for Long-term Immunosuppression

- 2 to 3 agents from

  - a calcineurin inhibitor (cyclosporine, tacrolimus);

  - an anti-metabolite (azathioprine, mycophenolic acid);

  - and sometimes a corticosteroid; wean as soon as possible

Advantage of Multiple Agents is Non-Overlapping Toxicity

Drug	Dose-Limiting Toxicity
Cyclosporine & tacrolimus	Nephrotoxicity, neurotoxicity, dyslipidemia
Sirolimus	Myelosuppression, hepatic toxicity dyslipidemia
Mycophenolate	GI irritation, myelosuppression

All these drugs increase the risk of infection and lymphoma

Immunosuppressive Drugs: Individual Toxicities

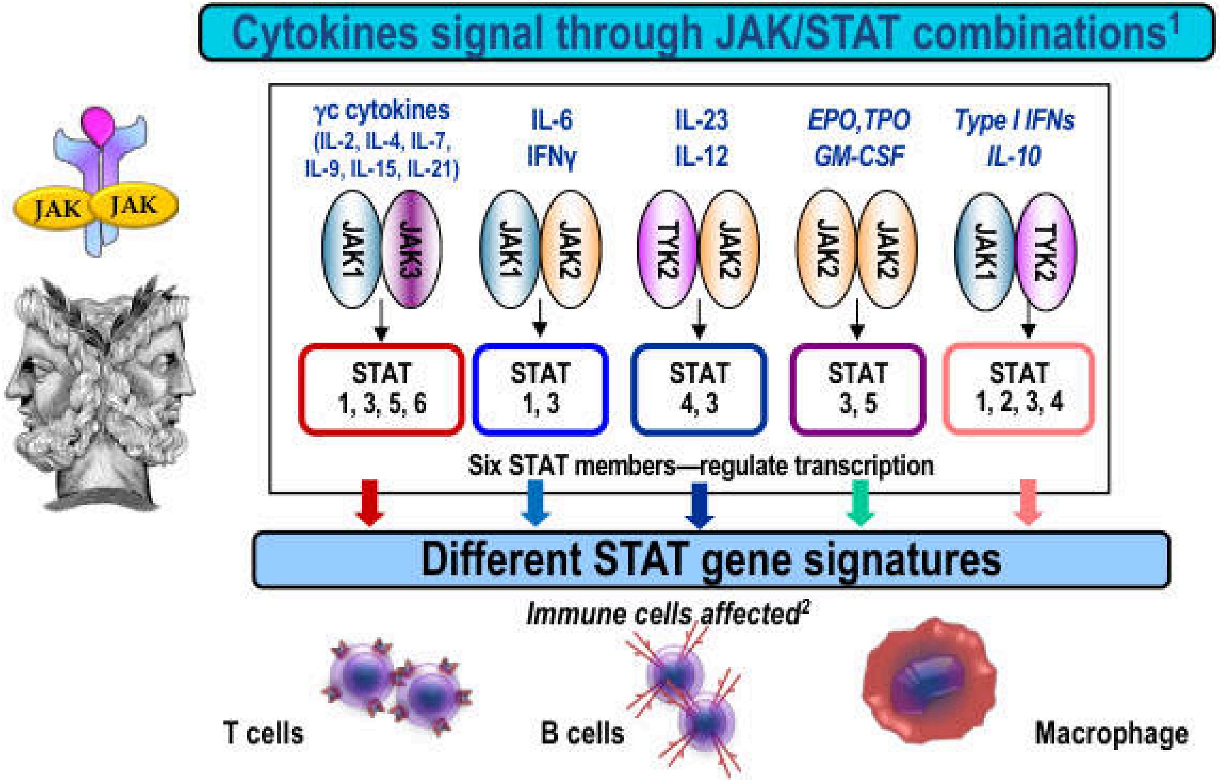
	Obesity	HTN	DM	Nephro	Neuro	Lipids	Marrow	GIT	Inf
Steroids	+++	+++	+++		+	+++			
CNIs									
Cyclosporin A		+++	+	+++	+++	++			
Tacrolimus		+++	++	+++	+++	+			
mTOR Inhibitors									
Sirolimus				++		+++	++		
Antimetabolites									
Azathioprine							+++		
Mycophenolate							+++	+++	
Antilymphocyte Antibodies									
ALG									++
OKT3									++
IL2R-Ab									

Inf = Infusion reaction

The Role of JAKiNib in Autoimmunity

- Immune homeostasis is the ability of the immune system to respond dynamically to the environment in such a way that the health and functional status of the subject is preserved and stabilised over time
- A complex cytokine network balances pro-inflammatory and anti-inflammatory effects. In RA, pro-inflammatory cytokines are predominant, but this is not constant
- A goal of therapy is to restore homeostasis without causing immune suppression
- Cytokines are implicated at each phase of RA pathogenesis
- Tofacitinib has a strong effect on multiple cytokines and the processes orchestrate

JAK/STAT Pathways Central to Cytokine Signaling



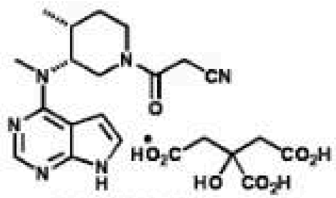
EPO=erithropoetin; GM-CSF=granulocyte/macrophage colony stimulating factor; IFN=interferon; IL=interleukin; JAK=Janus kinase; STAT=signal transducer and activator of transcription; TPO=thrombopoietin; TYK2=tyrosine kinase 2.

1. O'Sullivan LA, et al. *Mol Immunol.* 2007;44:2497-506; 2. Murray PJ. *J Immunol.* 2007;178:2623-2629. Figure adapted from Riese RJ, et al. *Best Pract Clin Res Rheumatol.* 2010;24:5113-526.

Structure and Pharmacology

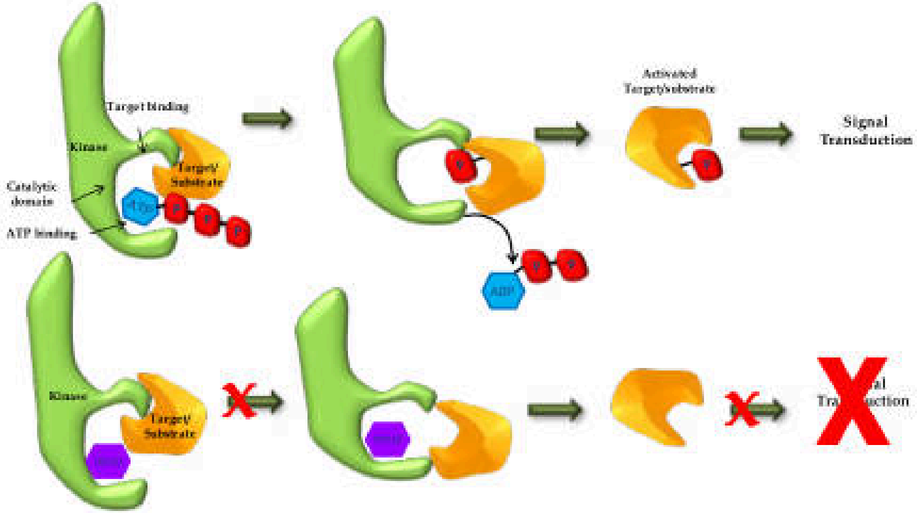
CP-690,550, is a synthetic small molecule designed as a competitive JAK inhibitor

- 300-fold smaller than etanercept
- A reversible competitor of the ATP site
- Rapid elimination with a $t_{1/2}$ of approximately 3 hrs
- Primary clearance mechanism of non-renal 70% (hepatic metabolism) and renal 30% excretion



CP-690550-10

- Molecular weight (citrate salt) is 504.5 Da



Toxicity of Jakinibs

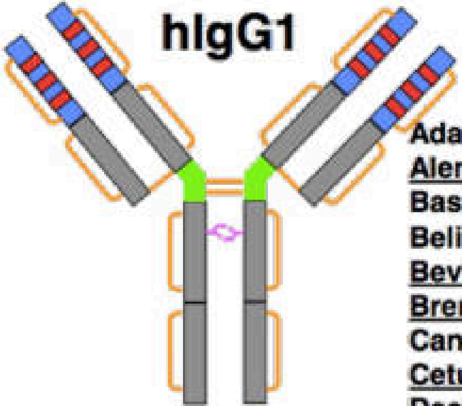
- Infection, few opportunistic infections in RA trials, Herpes zoster
- Cytopenias: anemia, leukopenia
  - Presumably related to Jak2 inhibition
- Tumors?
- Increased lipids
  - Lipid effect seen with tocilizumab (anti-IL-6R)
  - Mechanism? Significance?
- Increased creatinine, transaminases, GI perforations
  - Mechanism? Related to Jaks? Unrelated?

Antibody Naming

Prefix	Target			Antibody Source		Suffix	Some Examples
Variable	Non-tumor target	Viral	-vir-	-u-	Human	-mab	pali-vi-zu-mab (humanized antiviral Mab)
		Bacterial	-bac-				
		Immune	-lim-	-o-	Murine		Ada-lim-u-mab (human Mab against immune disease target)
		Infectious Lesions	-les-				
		Antifungal	-fung-	-a-	Rat		E-fung-u-mab (human antifungal Mab)
		Cardiovascular	-ci(r)-				
		Neurologic	-ne(r)-	-e-	Hamster		Bapi-neu-zu-mab (Humanized Mab against neurobiology target)
		Interleukins	-kin-				
		Musculoskeletal	-mul-	-i-	Primate		
		Bone	-os-				Uste-kin-u-mab (human anticytokine Mab)
		Toxin as target	-toxa-	-xi-	Chimeric		Den-os-u-mab (human antibone target Mab)
	Tumor Target	Colon	-col-				
		Melanoma	-mel-	-zu-	Humanized		Ab-ci-xi-mab (chimeric Mab against CV target)
		Mammary	-mar-				
		Testis	-got-	-axo-	Rat/murine hybrid		Ore-gov-o-mab (murine Mab for ovarian cancer)
		Ovary	-gov-				
		Prostate	-pr(o)-	-xizu-	Chimeric+ humanized		Adeca-tum-u-mab (human antibody against miscellaneous tumor target)
		Miscellaneous	-tu(m)-				

IgG1, 2 & 4 structures (Ch/Hz/Hu)

hIgG1

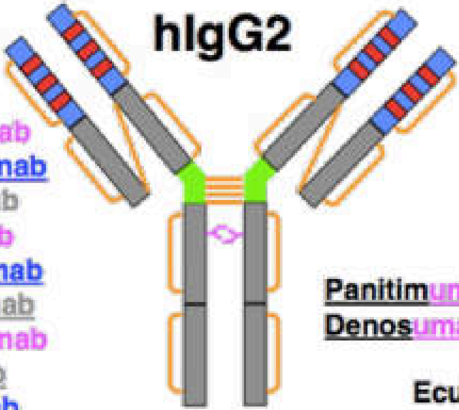


Adalimumab
Alemtuzumab
Basiliximab
Belimumab
Bevacizumab
Brentuximab
Canakinumab
Cetuximab
Daclizumab
Efalizumab
Golimumab
Infliximab
Ipilimumab
Nimotuzumab
Mogamulizumab
Ofatumumab
Omalizumab
Palivizumab
Rituximab
Tocilizumab
Trastuzumab
Ustekinumab

- 16 Cys-Cys (2 inter-HC)
- LC-Cys⁵- HC-Cys⁵

ADCC/ CDC ++

hIgG2

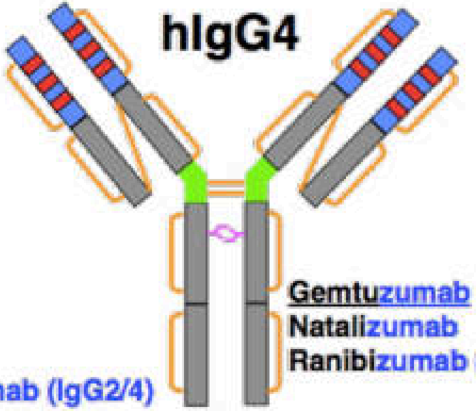


Panitimumab
Denosumab

Eculizumab (IgG2/4)

- 18 Cys-Cys (4 inter-HC)
- LC-Cys⁵- HC-Cys³ (+ isomers, dimers)

hIgG4



Gemtuzumab (IC)
Natalizumab
Ranibizumab (Fab)

- 16 Cys-Cys (2 inter-HC)
- LC-Cys⁵- HC-Cys³ (+ half-Abs, bispecifics)

ADCC/ CDC -

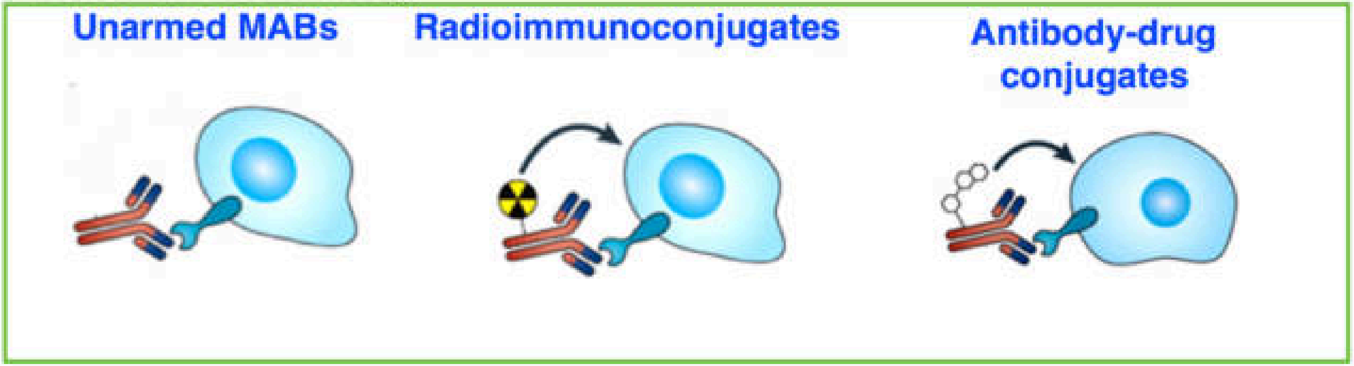
- Beck A et al, Nature Reviews Immunology, 2010
- Reichert J, mAbs, 2012

MAbs in Solid Organ Transplantation

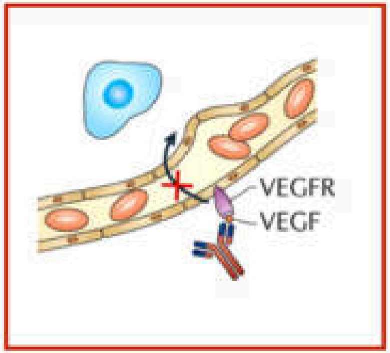
Monoclonal antibody	Molecular weight	Animal epitope	Molecular target	Target cells	Use
Alemtuzumab (Campath-1H®)	150 kDa	Murine/ human	CD52	Peripheral blood lymphocytes, natural killer cells, monocytes, macrophages, thymocytes	Induction Antibody-mediated rejection
Daclizumab (Zenapax®)	14.4 kDa	Murine/ human	CD25 alpha subunit	IL2-dependent T-lymphocyte activation	Induction
Basiliximab (Simulect®)	14.4 kDa	Murine/ human	CD25 alpha subunit	IL2-dependent T-lymphocyte activation	Induction
Muromonab-OKT3 (Orthoclone-OKT3®)	75 kDa	Murine	CD3	T lymphocytes (CD2, CD4, CD8)	Treatment of polyclonal antibody-resistant cellular-mediated rejection
Rituximab (Rituxan®)	145 kDa	Murine/ human	CD20	B lymphocytes	Desensitization Antibody-mediated rejection Focal segmental glomerulosclerosis
Belatacept (Nulojix®)	90 kDa	Humanized	CD80 and CD86	T lymphocytes	Maintenance immunosuppression
Eculizumab (Soliris®)	148 kDa	Murine/ human	C5	Block formation of membrane attack complex	Desensitization Antibody-mediated rejection Hemolytic uremic syndrome

Monoclonal Antibodies in Cancer

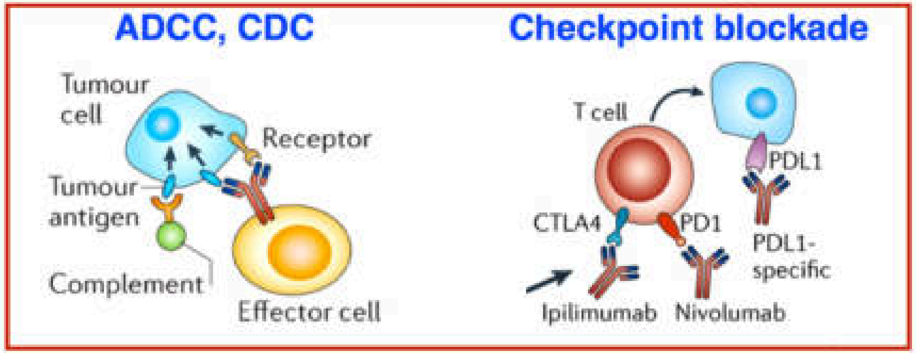
Direct Tumor Killing



Vascular Ablation



Immune-Mediated Tumor Cell Killing



Monoclonal Antibodies in Cancer

Generic name	Approval year	Origin	Target	Indications
Cluster of differentiation (CD) targeted				
Alemtuzumab	2001	Humanized	CD-52	B-cell CLL
Gemtuzumab	2000[a]	Humanized	CD-33	AML –compassionate use program only
Rituximab	1997	Chimeric	CD-20	Non-Hodgkin's lymphoma
Yttrium-90 (^{90}Y) ibritumomab tiuxetan	2002	Murine	CD-20	B-cell non-Hodgkin's lymphoma
Iodine-131 (^{131}I) tositumomab	2003	Murine	CD-20	Non-Hodgkin's lymphoma
Brentuximab vedotin	2011	Chimeric	CD-30	Hodgkin's lymphoma
Ofatumumab	2009	Human	CD-20	CLL
Epidermal growth factor receptor (EGFR) targeted				
Cetuximab	2004	Chimeric	EGFR	Colorectal cancer, SCCHN
Panitumumab	2006	Human	EGFR	Colorectal cancer
Trastuzumab	1998	Humanized	HER2/neu	Breast cancer, gastric cancer
Vascular endothelial growth factor (VEGF) targeted				
Bevacizumab	2004	Humanized	VEGF	NSCLC, colorectal cancer, glioblastoma multiforme, metastatic renal cell carcinoma
Receptor activator of nuclear factor kappa B ligand (RANKL)				
Denosumab	2010	Human	RANKL	Cancer metastatic to the bone
Cytotoxic T-lymphocyte antigen-4 (CTLA-4)				
Ipilimumab	2011	Human	CTLA-4	Metastatic melanoma

CLL chronic lymphocytic leukemia, *AML* acute myeloid leukemia, *VEGF* vascular endothelial growth factor, *NSCLC* non-small cell lung cancer, *EGFR* endothelial growth factor receptor, *SCCHN* squamous cell carcinoma of the head and neck

[a]Approval subsequently denied.

MAbs in Inflammatory Diseases

Product	Target	Indication	Pharmacokinetics	Recommended dose regimen
Abatacept (Orencia®)	CTLA-4	RA	Abatacept exhibits linear pharmacokinetics with mean terminal half-life being 13.1–14.3 days. The mean (range) systemic CL was estimated to be 0.22 (0.13–0.47) mL/h/kg. The bioavailability of SC abatacept was 78.6 %. Following IV infusions of 10 mg/kg q4w in adult RA patients, the mean (range) steady-state peak and trough concentrations were 295 (171–398) and 24 (1–66) μg/mL, respectively. Following SC injections of 125 mg weekly, the mean (range) steady-state peak and trough concentrations were 48.1 (9.8–132.4) and 32.5 (6.6–113.8) μg/mL, respectively.	*IV infusion for RA*: For adult RA patients weighing <60 kg, the recommended dose of IV abatacept is 500 mg. For patients weighing 60–100 kg, a 750-mg dose should be administered. For patients weighing >100 kg, a 1,000-mg dose should be given. Abatacept should be administered as IV infusion at weeks 0, 2, and 4 and q4w thereafter. *SC injection for RA*: 125 mg should be administered weekly following an initial IV loading dose (according to body weight categories described above).
Alefacept (Amevive®)	CD2	PsO	The mean clearance of alefacept was 0.25 mL/h/kg with mean terminal half-life being approximately 270 h. The mean bioavailability of alefacept following an intramuscular injection was 63 %.	Alefacept should be administered as intramuscular injection. *PsO*: The recommended dose of alefacept is 15 mg once weekly for 12 weeks. The CD4+ T-lymphocyte count should be measured before initial dosing.
Adalimumab (Humira®)	TNFα	RA, PsA, AS, PsO, and CD	Adalimumab exhibits linear pharmacokinetics with mean terminal half-life being about 2 weeks. The systemic clearance of adalimumab is approximately 12 mL/h. The mean bioavailability of SC adalimumab was 64 %. Following SC administration of 40 mg q2w in adult RA patients, the mean steady-state trough concentrations were about 5 and 8–9 μg/mL without and with concomitant methotrexate, respectively. Methotrexate reduced adalimumab apparent clearance after single and multiple dosing by 29 % and 44 %, respectively, in patients with RA.	Adalimumab should be administered as SC injection. *RA, PsA, and AS*: The recommended dose of adalimumab is 40 mg SC q2w. *PsO*: The recommended dose is 80 mg initial dose followed by 40 mg SC q2w. *CD*: The recommended dose is 160 mg initial dose, 80 mg dose 2 weeks later, followed by 40 mg SC q2w.
Belimumab (Benlysta®)	BLyS	SLE	The median systemic clearance of belimumab was estimated to be 215 mL/day with a median terminal half-life being 19.4 days.	Belimumab should be administered as IV infusion. *SLE*: The recommended dose of belimumab is 10 mg/kg q2w for the first three doses, followed by 10 mg/kg q4w thereafter.
Canakinumab (Ilaris®)	IL-1	CAPS	Canakinumab exhibits linear pharmacokinetics with mean terminal half-life being 26 days. The median systemic clearance for a CAPS patient weighing 70 kg was estimated to be 0.174 L/day. The mean bioavailability of SC canakinumab was 70 %.	Canakinumab should be administered as SC injection. *CAPS*: The recommended dose of canakinumab is 150 mg SC q8w.
Certolizumab pegol (Cimzia®)	TNFα	RA and CD	Certolizumab pegol exhibits linear pharmacokinetics with mean terminal half-life being about 14 days. The systemic clearance following IV administration to healthy subjects ranged from 9.21 to 14.38 mL/h. The mean bioavailability of SC certolizumab pegol was 80 %.	Certolizumab pegol should be administered as SC injection. *RA*: The recommended dose of certolizumab pegol is 400 mg initially and 400 mg at weeks 2 and 4, followed by a lower maintenance dosing regimen of 200 mg q2w thereafter. For maintenance dosing, 400 mg q4w can also be considered. *CD*: The recommended dose of certolizumab pegol is 400 mg initially and 400 mg at weeks 2 and 4. For patients achieving clinical response, the recommended maintenance dose regimen is 400 mg q4w.

MAbs in Inflammatory Diseases

Etanercept (Enbrel®)	TNFα	RA, PsA, AS, and PsO	Etanercept exhibits linear pharmacokinetics with mean terminal half-life being 102 ± 30 h. The mean clearance was 160 ± 80 mL/h. Following SC administration of 50 mg etanercept weekly in adult RA patients, the mean ± SD steady-state peak and trough concentrations were 2.4 ± 1.5 and 1.2 ± 0.7 μg/mL.	Etanercept should be administered as SC injection. *RA, PsA, and AS*: The recommended dose of etanercept is 50 mg SC weekly. *PsO*: The recommended dose of etanercept is 50 mg twice weekly for 3 months, followed by 50 mg SC weekly.
Golimumab (Simponi®)	TNFα	RA, PsA, AS	Golimumab exhibits linear pharmacokinetics with mean terminal half-life being about 2 weeks. The mean systemic clearance was estimated to be 4.9–6.7 mL/day/kg. The mean bioavailability of SC golimumab was 53 %. Following SC administration of 50 mg q4w with concomitant methotrexate in adult RA patients, the mean steady-state trough concentrations were 0.4–0.6 μg/mL. Concomitant use of methotrexate reduced the apparent clearance of golimumab by approximately 36 %.	Golimumab should be administered as SC injection. *RA, PsA, and AS*: The recommended dose of golimumab is 50 mg SC monthly.
Infliximab (Remicade®)	TNFα	RA, PsA, AS, PsO, CD, UC	Infliximab exhibits linear pharmacokinetics with median terminal half-life being 7.7–9.5 days. Following IV infusions of a maintenance dose of 3–10 mg/kg q8w, median steady-state serum infliximab concentrations ranged from approximately 0.5–6 μg/mL.	Infliximab should be administered as IV infusion. *RA*: The recommended dose of infliximab is 3 mg/kg at 0, 2, and 6 weeks, followed by 3 mg/kg q8w thereafter. Some patients may benefit from increasing the dose up to 10 mg/kg or treating as often as q4w. *PsA and PsO*: The recommended dose of infliximab is 5 mg/kg at 0, 2, and 6 weeks, followed by 5 mg/kg q8w thereafter. *AS*: The recommended dose of infliximab is 5 mg/kg at 0, 2, and 6 weeks, followed by 5 mg/kg q6w thereafter. *CD and UC*: The recommended dose of infliximab is 5 mg/kg at 0, 2, and 6 weeks, followed by 5 mg/kg q8w thereafter. Some CD patients who initially respond to treatment may benefit from increasing the dose to 10 mg/kg if they later lose their response.
Natalizumab (Tysabri®)	Integrins	CD and MS	In patients with CD, the systemic clearance of natalizumab was 22 ± 22 mL/h with mean half-life being 10 ± 7 days. The mean steady-state trough concentration was 10 ± 9 μg/mL after q4w dosing. In patients with MS, the systemic clearance of natalizumab was 16 ± 5 mL/h with mean half-life being 11 ± 4 days. The mean steady-state trough concentrations ranged from 23 to 29 μg/mL after q4w dosing.	Natalizumab should be administered as IV infusion. *CD and MS*: The recommended dose of natalizumab is 300 mg q4w.
Omalizumab (Xolair®)	IgE	Asthma	Omalizumab exhibits target-mediated nonlinear pharmacokinetics. The mean apparent clearance of omalizumab was estimated to be 2.4 ± 1.1 mL/kg/day in asthma patients with a mean terminal half-life of 26 days. The mean bioavailability of SC omalizumab was 62 %.	Omalizumab should be administered as SC injection. *Asthma*: The recommended dose of omalizumab is 150–375 mg SC q2w or q4w, with dosage and dosing frequency determined by pretreatment serum IgE levels and body weight.

MAbs in Inflammatory Diseases

Product	Target	Indication	Pharmacokinetics	Recommended dose regimen
Rilonacept (Arcalyst)	IL-1	CAPS	Following weekly SC doses of 160 mg in patients with CAPS, serum rilonacept concentration appeared to reach steady state by 6 weeks with average trough levels being approximately 24 μg/mL.	Rilonacept should be administered as SC injection. *CAPS*: The recommended dose of rilonacept is 320 mg SC as a loading dose, followed by a maintenance dose regime of 160 mg SC weekly.
Rituximab (Rituxan®)	CD20	RA	The estimated clearance of rituximab was 0.335 L/day in patients with RA with mean terminal elimination half-life being 18.0 days.	Rituximab should be administered as IV infusion. Treatment courses of rituximab should be administered every 24 weeks or based on clinical evaluation, but no more frequent than every 16 weeks. *RA*: The recommended dose of each course for the treatment of RA is two 1,000 mg intravenous infusions separated by 2 weeks.
Tocilizumab (Actemra®)	IL-6 receptor	RA	Tocilizumab exhibits nonlinear pharmacokinetics with mean apparent terminal half-life being 151 ± 59 h after a single dose of 10 mg/kg in RA patients. The mean clearance was 0.29 ± 0.10 mL/h/kg at a dose level of 10 mg/kg. Following repeated dosing of 8 mg/kg IV q4w in adult RA patients, the model-predicted mean (SD) peak and trough concentrations at steady state were 183 ± 85.6 and 9.7 ± 10.5 μg/mL, respectively.	Tocilizumab should be administered as IV infusion. *RA*: The recommended starting dose of tocilizumab is 4 mg/kg IV q4w, followed by an increase to 8 mg/kg q4w based on clinical response.
Ustekinumab (Stelara®)	IL12/IL23	PsO	Ustekinumab exhibits linear pharmacokinetics with mean terminal half-life being 14.9–45.6 days. The mean systemic clearance in psoriasis patients ranged from 1.90 ± 0.28 to 2.22 ± 0.63 mL/day/kg. Following SC administration of 45 or 90 mg q12w in psoriasis patients, the mean steady-state trough concentrations were 0.31 ± 0.33 and 0.64 ± 0.64 μg/mL, respectively.	Ustekinumab should be administered as SC injection. *PsO*: The recommended dose of ustekinumab is 45 mg initially and 4 weeks later, followed by 45 mg q12w for patients weighing ≤100 kg; the recommended dose for patients weighing >100 kg is 90 mg initially and 4 weeks later, followed by 90 mg q12w.

Abbreviations: *AS* ankylosing spondylitis, *BLyS* B-lymphocyte stimulator, *CAPS* cryopyrin-associated periodic syndromes, *CD* Crohn's disease, *CD2* T-lymphocyte antigen CD2, *CD20* B-lymphocyte antigen CD20, *CTLA-4* cytotoxic T-lymphocyte-associated antigen 4, *IgE* immunoglobulin E, *IL* interleukin, *IV* intravenous, *MS* multiple sclerosis, *PsA* psoriatic arthritis, *PsO* plaque psoriasis, *q12w* every 12 weeks, *q2w* every 2 weeks, *q4w* every 4 weeks, *q8w* every 8 weeks, *RA* rheumatoid arthritis, *SC* subcutaneous, *SLE* systemic lupus erythematosus, *TNFα* tumor necrosis factor alpha, *UC* ulcerative colitis

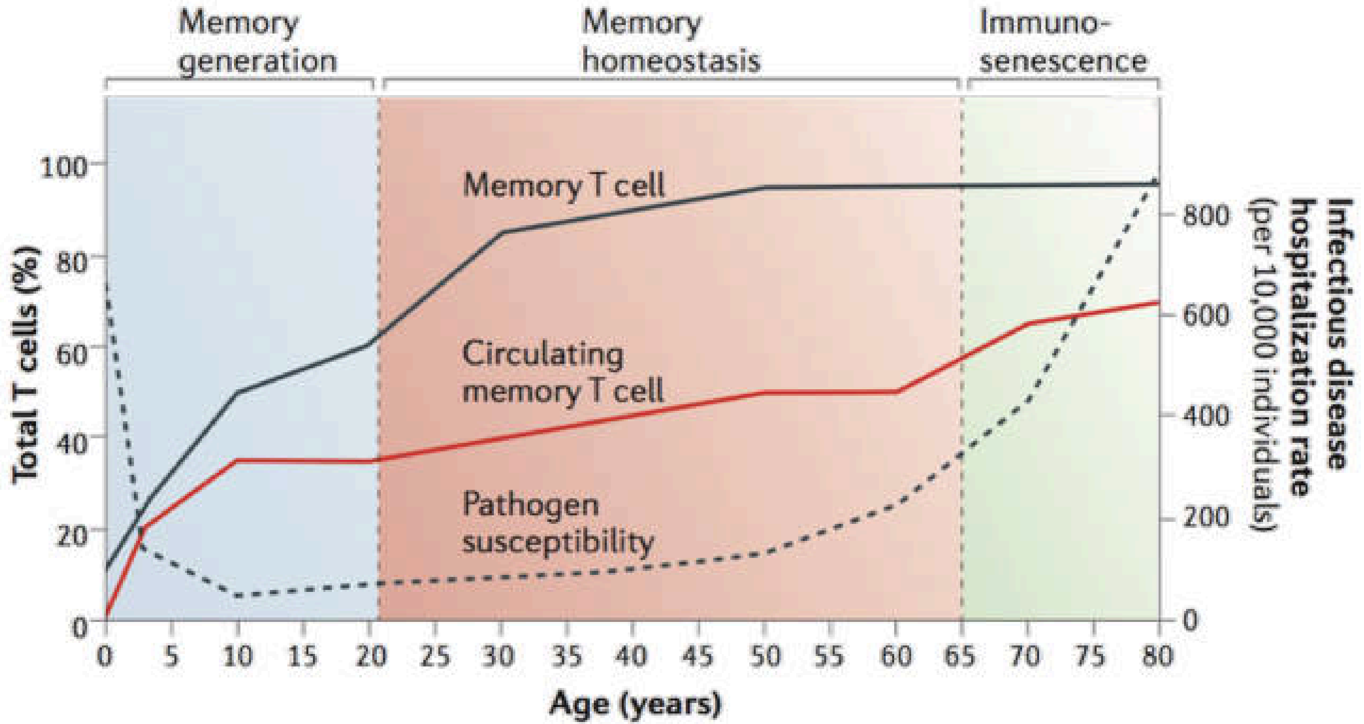
Immunostimulators

There are two main categories of immunostimulants:

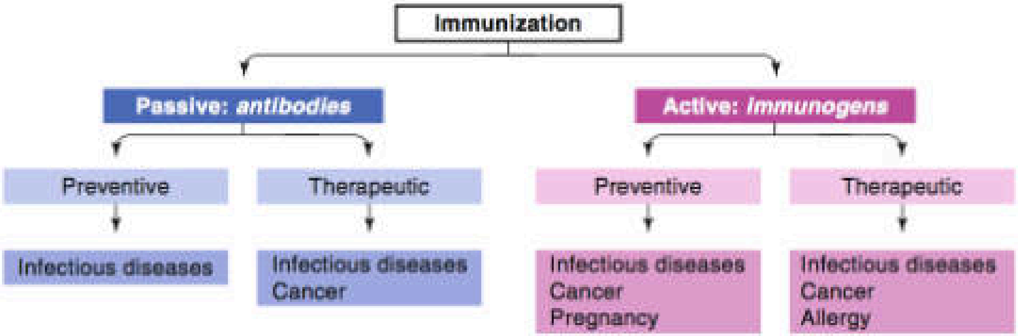
- **Specific immunostimulants** provide antigenic specificity in immune response, such as **vaccines** or any antigen
- **Non-specific immunostimulants** act irrespective of antigenic specificity to augment immune response of other antigen or stimulate components of the immune system

- Vaccine
- Immunoglobulin
- Myeloid growth factors
- Interferons
- Imiquimod
- Others

Memory T cell frequency, pathogen susceptibility and mortality



Nature Reviews Immunology 14, 24–35 (2014)



Categories of classical vaccines and vaccines obtained by modern technologies

Category	Technology	Live/nonliving	Characteristics
Attenuated vaccines	Classical	Live	Bacteria or viruses attenuated in culture; empirically developed
Inactivated vaccines	Classical	Nonliving	Heat-inactivated or chemically inactivated bacteria or viruses; empirically developed
Subunit vaccines	Classical	Nonliving	Extracts of pathogens; combination of purified proteins with killed suspension; purified single components (proteins, polysaccharides); combination of purified components with adjuvant; purified components in a suitable presentation form; polysaccharide-protein conjugates
Genetically improved live vaccines	Modern	Live	Genetically attenuated microorganisms; live viral or bacterial vectors
Genetically improved subunit vaccines	Modern	Nonliving	Proteins expressed in host cells; recombinant protein/peptide vaccines
Recombinant subunit vaccines identified by reverse vaccinology	Modern	Nonliving	Recombinant antigenic proteins obtained from the genomic sequence of the pathogen
Synthetic peptide-based vaccines	Modern	Nonliving	Linear or cyclic peptides; multiple antigen peptides; peptide-protein conjugates
Nucleic acid-based vaccines	Modern	Nonliving	DNA or mRNA coding for antigen

Vaccines	Vaccine type	Serum IgG	Mucosal IgG	Mucosal IgA	T cells
Diphtheria toxoid	toxoid	++	(+)		
Hepatitis A	killed	++			
Hepatitis B (HbsAg)	protein	++			
Hib PS	PS	++	(+)		
Hib glycoconjugates	PS-protein	++	++		
Influenza	killed, subunit	++	(+)		
Influenza intranasal	live attenuated	++	+	+	+ ($CD8^+$)
Measles	live attenuated	++			+ ($CD8^+$)
Meningococcal PS	PS	++	(+)		
Meningococcal conjugates	PS-protein	++	++		
Mumps	live attenuated	++			
Papillomavirus	VLPs	++	++		
Pertussis, whole cell	killed	++			
Pertussis, acellular	protein	++			+?($CD4^+$)
Pneumococcal PS	PS	++	(+)		
Pneumococcal conjugates	PS-protein	++	++		
Polio Sabin	live attenuated	++	++	++	
Polio Salk	killed	++	+		
Rabies	killed	++			
Rotavirus	live attenuated			++	
Rubella	live attenuated	++			
Tetanus toxoid	toxoid	++			
Tuberculosis (BCG)	live mycob				++($CD4^+$)
Typhoid PS	PS	+	(+)		
Varicella	live attenuated	++			+?($CD4^+$)
Yellow Fever	live attenuated	++			

PS : polysaccharide
VLP : virus-like-particle

ตารางการสร้างเสริมภูมิคุ้มกันโรคในเด็กและวัยรุ่นไทย แนะนำโดย สมาคมโรคติดเชื้อในเด็กแห่งประเทศไทย 2569

วัคซีนจำเป็นที่ต้องให้กับเด็กทุกคน

วัคซีน / อายุ	แรกเกิด	1 เดือน	2 เดือน	4 เดือน	6 เดือน	9-12 เดือน	18 เดือน	2-2½ ปี	4-6 ปี	11-12 ปี
บีซีจี[1] (BCG)	BCG									
ตับอักเสบบี[2] (HB)	HB1	(HB2)	DTwP-HB-Hib-1	DTwP-HB-Hib-2	DTwP-HB-Hib-3					
คอตีบ-บาดทะยัก-ไอกรนชนิดทั้งเซลล์[3] (DTwP)							DTwP กระตุ้น 1		DTwP กระตุ้น 2	Td และ ทุก 10 ปี
ฮิบ[4] (Hib)										
โปลิโอ[5] ชนิดฉีด (IPV) และกิน (OPV)			IPV1	IPV2	OPV1		OPV กระตุ้น 1		OPV กระตุ้น 2	
โรต้า[6] (Rota)			Rota1	Rota2	(Rota3)					
หัด-คางทูม-หัดเยอรมัน[7] (MMR)						MMR1	MMR2			
ไข้สมองอักเสบเจอีชนิดเชื้อมีชีวิต[8] (live JE)						JE1		JE2		
ไข้หวัดใหญ่[9] (Influenza)					Influenza ให้ปีละครั้ง (การให้ครั้งแรกในเด็กอายุต่ำกว่า 9 ปี ให้ 2 เข็ม ห่างกัน 1 เดือน)					
เอชพีวี[10] (HPV)										เด็กหญิงประถม 5 ตามแผนฯ ของกระทรวงสาธารณสุข

วัคซีนอื่น ๆ หรือภูมิคุ้มกันสำเร็จรูป ที่อาจให้เสริม หรือทดแทน

วัคซีน / อายุ	แรกเกิด	2 เดือน	4 เดือน	6 เดือน	12-15 เดือน	18 เดือน	2-2½ ปี	4 ปี	6 ปี	9 ปี	11-12 ปี	18 ปี
คอตีบ-บาดทะยัก-ไอกรน ชนิดไร้เซลล์[3] หรือ ชนิดทั้งเซลล์ (DTaP/DTwP, Tdap หรือ TdaP) ตับอักเสบบี โปลิโอ[5] ชนิดฉีด (IPV) ฮิบ[4] (Hib)		DTaP/DTwP-HB-IPV-Hib1	DTaP/DTwP-(HB)-IPV-Hib2	DTaP/DTwP-HB-IPV-Hib3		DTaP-IPV-(Hib4) กระตุ้น 1		DTaP-IPV หรือ Tdap-IPV หรือ TdaP กระตุ้น 2			Tdap หรือ TdaP ต่อไป Td หรือ Tdap/TdaP ทุก 10 ปี	
นิวโมคอคคัสชนิดคอนจูเกต[11] (PCV)		PCV1	PCV2	(PCV3)	PCV4							
ไข้สมองอักเสบเจอีชนิดเชื้อไม่มีชีวิต[8] (Inactivated JE)				JE1, JE2 ห่างกัน 4 สัปดาห์ และ JE3 อีก 1 ปี								
อีวี 71[12] (EV71)				EV71 2 เข็ม ห่างกัน 1 เดือน								
ตับอักเสบเอ[13] (HAV)					HAV ชนิดเชื้อไม่มีชีวิต ให้ 2 ครั้ง ห่างกัน 6-12 เดือน							
อีสุกอีใส[14] (VZV) หรือวัคซีนรวม หัด-คางทูม-หัดเยอรมัน-อีสุกอีใส (MMRV)					VZV1 (หรือ MMRV1)	VZV2 (หรือ MMRV2)						
วัคซีนไข้หวัดใหญ่[9] (Influenza)				Influenza ให้ปีละครั้ง (การให้ครั้งแรกในเด็กอายุต่ำกว่า 9 ปี ให้ 2 ครั้ง ห่างกัน 1 เดือน) ชนิดเชื้อไม่มีชีวิตฉีดเมื่ออายุ 6 เดือนขึ้นไป ชนิดเชื้อมีชีวิตพ่นจมูกเมื่ออายุ 2 ปีขึ้นไป								
เอชพีวี[10] (HPV)										HPV 2 เข็ม ห่างกัน 6-12 เดือน		
ไข้เลือดออก[15] (DEN)								DEN 2 เข็ม ห่างกัน 3 เดือน				
พิษสุนัขบ้า[16] (Rabies) ก่อนการสัมผัสโรค	2 ครั้งห่างกันอย่างน้อย 7 วัน											
วัคซีนไข้กาฬหลังแอ่นซีโรกรุ๊ปบี[17] (MenB)		MenB1	MenB2		MenB3							
โควิด-19 (COVID-19)				ดูคำแนะนำในการฉีดตามคำแนะนำของกรมควบคุมโรค กระทรวงสาธารณสุข และราชวิทยาลัยกุมารแพทย์แห่งประเทศไทย								
ภูมิคุ้มกันสำเร็จรูปป้องกันอาร์เอสวีชนิดรุนแรง[18]	แนะนำ 1 เข็มในเด็กอายุน้อยกว่า 1 ปี (ดูคำแนะนำในการฉีดตามคำแนะนำของราชวิทยาลัยกุมารแพทย์แห่งประเทศไทย)											

ฉบับวันที่ 3 กุมภาพันธ์ 2569

Vaccine Notification

- Route of administration
- Combined vaccines ex. DTP+ Hib+ HBV+IPV
- Timing and spacing of vaccines
- Adverse reactions following vaccination
- Contraindications and precautions
- Special population
  - Pregnant women
  - Immunocompromised person

Determinants of Optimal Immunization Schedules

- Immunological
  - Minimum age at which vaccine elicit a immune response
  - Number of doses required
  - Interval between doses, if multiple doses are required
- Epidemiological
  - Susceptibility for infection and disease
  - Disease severity and mortality
- Programmatic
  - Opportunity to deliver with other scheduled interventions
  - Increase coverage by limiting the required contacts
- Safety

Immunological Determinants: Vaccine and Antigen type

- Live versus non-live vaccines (Higher Ab, longer duration)
  - Higher intensity innate response with live vaccines
  - Higher antigen content following replication and prolonged antigen persistence with live vaccines
  - Use of adjuvants enhance response to non-live vaccines
- Antigen type
  - Polysaccharide antigen --- T-cell independent
    - Lower Ab response of shorter duration and no booster response
  - Protein antigens
    - Can vary with Ag (e.g. TT >DT) and Ag dose

Immunological Determinants: # of doses

- Number of doses required vary by vaccine
  - In general, live vaccines induce immunity with a single dose and inactivated vaccines require multiple doses
  - Some live vaccines only induce immunity in a relatively small proportion of vaccines, requiring multiple doses to induce immunity in an optimal % of population, e.g. OPV
- Number of doses required may also vary by age
  - More doses of conjugate vaccines required in young infants
- Duration of immunity and requirement for additional doses
  - Either to boost or to re-induce immunity (for T-cell independent antigens)
  - Non-live vaccines generally require booster doses for long term immunity
  - Since vaccines seldom produce sustained mucosal immunity, natural boosting from mucosal infection occurs frequently

Immunological Determinants: Age & Route of Administration

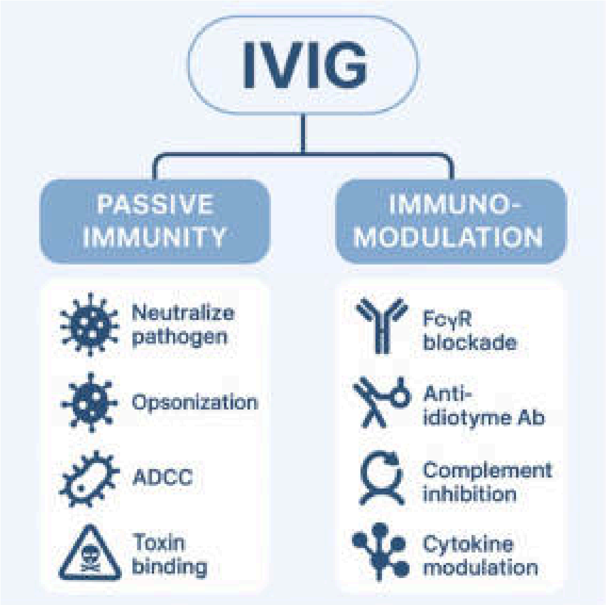
- Age
  - Maturity of the immune system
    - Inability to respond to PS antigen
    - Greater number of doses for primary response (PS-conjugate vaccines)
    - Inhibitory effect of maternal Ab
- Route of administration
  - Most non-live vaccines have to be given IM or ID because of presence of adequate number of APCs
  - Mucosal immunization with non-live vaccines is difficult
    - exception oral cholera vaccine (killed +/- B subunit)
  - Route of administration less of an issue with live vaccines
    - Organisms should be able to replicate at site of administration
    - Following replication organisms disseminate and induce generalized immune response

Balance between Immunological and Epidemiological Determinants

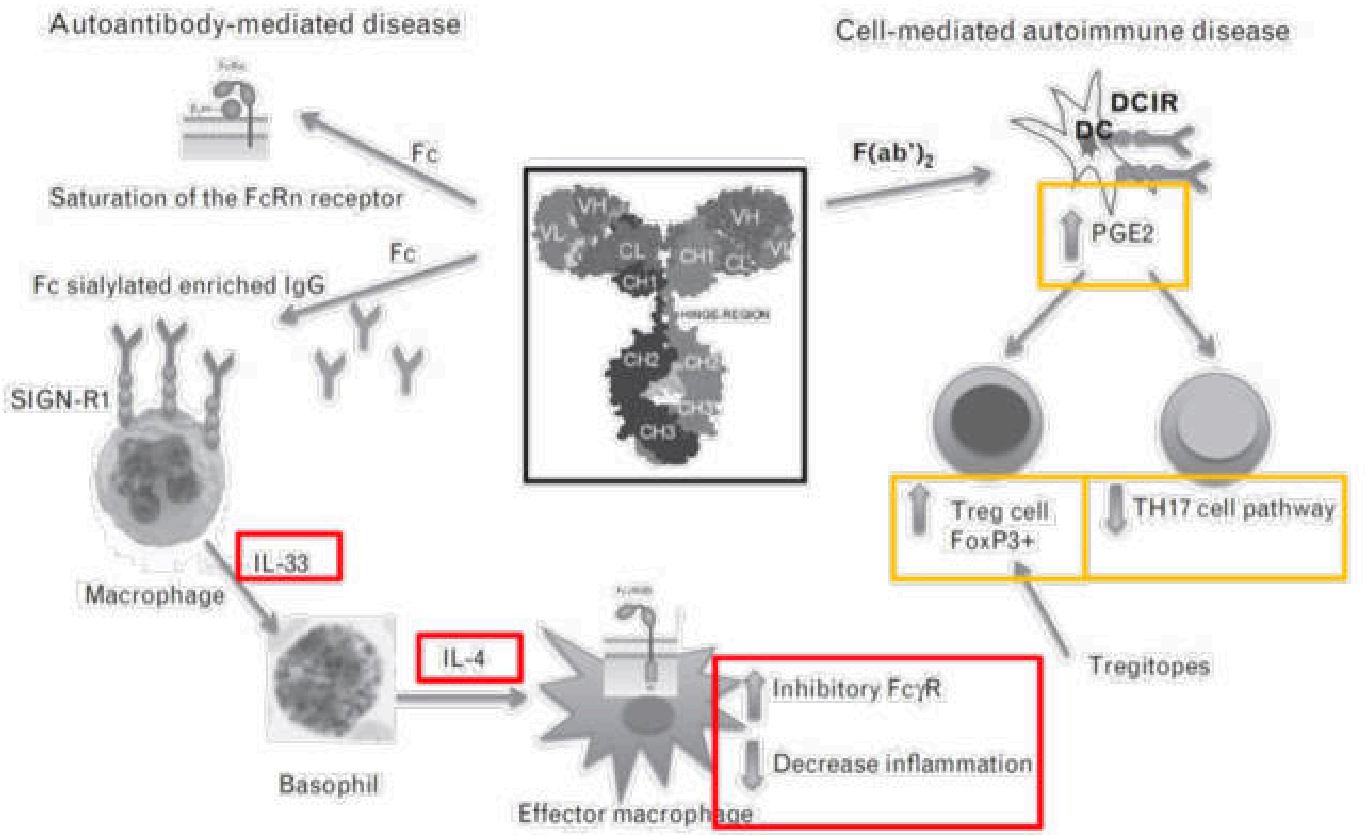
- Aim for achieving protective immune response prior to the age when children are most vulnerable
  - Balance between inducing reasonable protection prior to vulnerable age versus inducing optimal immune response
  - Starting late might induce a higher response, but miss the vulnerable age
  - Wider intervals between doses gives a better response, but delays induction of immunity, leaving children vulnerable in a crucial period in their life
- The disease epidemiology varies in different populations
  - A schedule that is used in one population is not the best for another, need to individualize and tailor to suit local needs

Human Normal Immunoglobulin

- Intravenous immunoglobulin G (IVIG)
- Polyclonal human immunoglobulin G
- No specific antigen epitope
- Indications
  - Allogeneic bone marrow transplant
  - Chronic lymphocytic leukemia
  - Idiopathic thrombocytopenic purpura
  - Pediatric HIV
  - Primary immunodeficiency
  - Kawasaki disease
  - Kidney transplant with a high antibody recipient or with an ABO incompatible donor
  - Other Off-label indications



Modulating Activities of IVIG



Ballow M. Curr Opin Allergy Clin Immunol 2014. 14:509–515

Intravenous Immunoglobulin

Mechanism	Description
FcγR blockade	Reduces phagocytosis of antibody-coated cells
FcγRIIB upregulation	Inhibits B-cell and macrophage activation
Anti-idiotypic antibodies	Neutralize pathogenic autoantibodies
Complement inhibition	Prevents complement-mediated tissue injury
Cytokine modulation	Shifts cytokine profile toward anti-inflammatory
T/B-cell regulation	Promotes Tregs, suppresses Th1/Th17 & autoreactive B cells
Pathogen neutralization	Binds infectious agents directly
ADCC & opsonization	Enhances clearance by immune effector cells

Intravenous Immunoglobulin

Immunomodulatory Mechanisms

Used in autoimmune and inflammatory diseases (e.g., ITP, Kawasaki disease, Myasthenia gravis, GBS)

MOA

- **Fc Receptor Modulation**
  - **FcγR Blockade**: IVIG saturates activating FcγRs on macrophages → reduces phagocytosis of opsonized cells (e.g., platelets in ITP)
  - **Upregulation of FcγRIIB (Inhibitory Fc receptor)** on B cells → dampens B-cell activation and autoantibody production
- **Anti-idiotypic Antibodies**
  - IVIG contains **anti-idiotype antibodies** that neutralize pathogenic autoantibodies (e.g., anti-AChR in MG, anti-platelet in ITP)
- **Complement Inhibition**
  - Binds C3b and C4b, inhibiting classical complement cascade activation → protects tissues from complement-mediated injury

Intravenous Immunoglobulin

MOA

- **Cytokine Modulation**
  - **Suppresses pro-inflammatory cytokines** (e.g., TNF-α, IL-1β, IL-6, IFN-γ)
  - **Enhances anti-inflammatory cytokines** (e.g., IL-10)
  - Alters monocyte/macrophage cytokine profiles
- **T-Cell and Dendritic Cell Regulation**
  - **Reduces Th1 and Th17 cells**, promotes **Tregs** (regulatory T cells)
  - Inhibits dendritic cell maturation → less antigen presentation → dampened T-cell activation
- **Inhibition of B-cell Maturation and Autoantibody Production**
  - Indirect suppression of autoreactive B cells via FcγRIIB signaling, idiotype neutralization, and cytokine effects
- **Scavenging of Pathogenic Autoantibodies**
  - IVIG binds soluble autoantibodies → forms immune complexes cleared by RES system

Intravenous Immunoglobulin

Passive Immunity Mechanisms Used in immunodeficiencies or infectious risk prevention (e.g., CVID, pediatric HIV)

MOA

- **Neutralization of Pathogens**
  - Preformed IgG against viruses, bacteria, and toxins neutralizes infection (e.g., CMV, RSV, measles, tetanus)
- **Opsonization**
  - IgG binds pathogens, enhances recognition by phagocytes → facilitates clearance
- **Antibody-Dependent Cellular Cytotoxicity (ADCC)**
  - IVIG-coated cells are targeted by **NK cells** and macrophages for destruction via FcγR engagement
- **Toxin Binding**
  - IVIG neutralizes bacterial superantigens and exotoxins (e.g., in streptococcal toxic shock)

Additional Observed Effects

- Modulation of gene expression in immune cells (e.g., downregulation of inflammatory transcription factors)
- MicroRNA regulation: certain miRNAs that regulate cytokine production are influenced by IVIG
- Neonatal Fc receptor (FcRn) saturation: increases catabolism of pathogenic IgG by reducing recycling

Rabies Immunoglobulin

- Source
  - Equine rabies immunoglobulin (ERIG)
  - Human rabies immunoglobulin (HRIG)
- Co-administration with rabies vaccine
- Indications – Single or multiple transdermal bites or scratches, licks on broken skin
- Administration – Around the bite sites

Anti-D Immunoglobulin

- Human IgG against Rh_0 (D) antigen of the red cell
- Use for prevention of Rh hemolytic disease on newborn
  - Rh_0 (D) –ve pregnant women with Rh_0 (D) +ve baby
  - Rh_0 (D) –ve pregnant women who had miscarriage, ectopic pregnancy, or abortion
- Administration at time of birth
- Immunoglobulin is administered to the mother
- Immunosuppressor

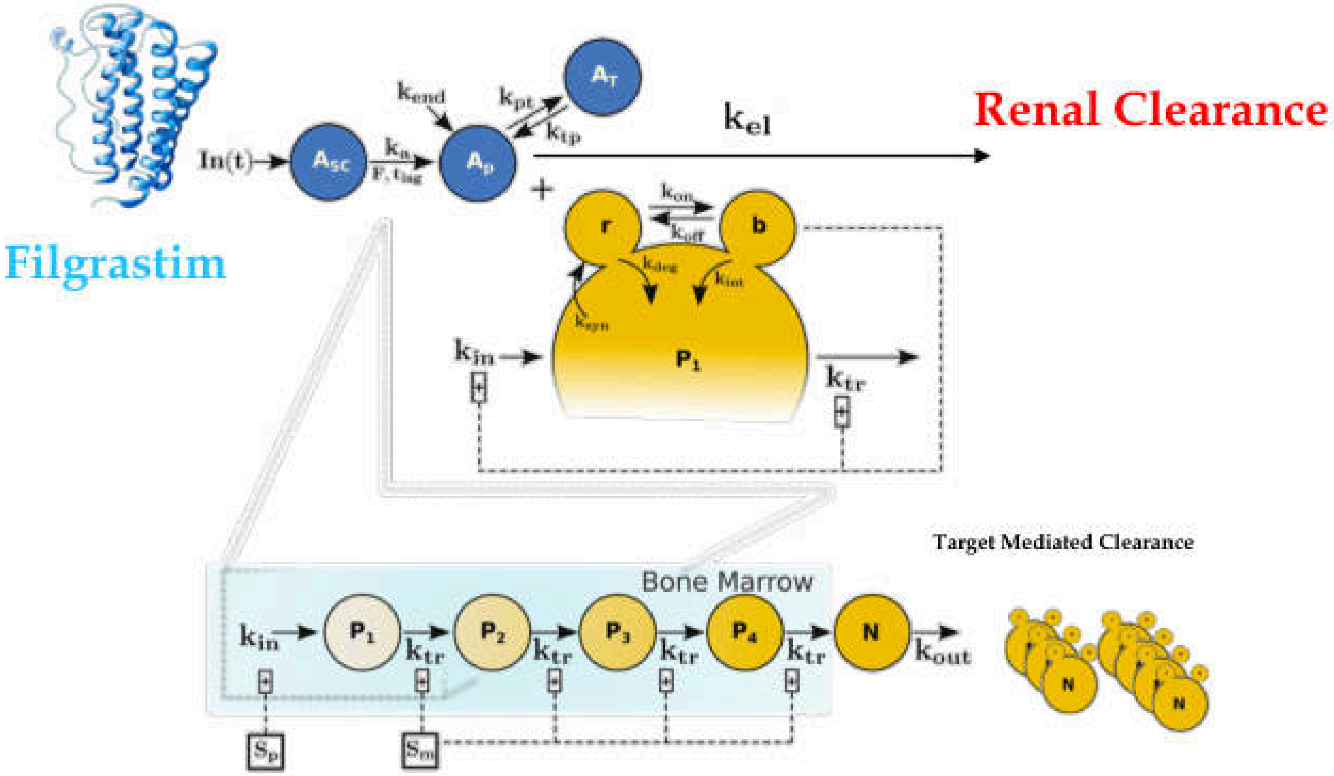
Other Immunoglobulins

- Human tetanus immunoglobulin
- Human cytomegalovirus immunoglobulin
- Anti-snake venom immunoglobulin
- Anti-human thymocyte immunoglobulin

Myeloid Growth Factors

- Activate JAK/STAT receptor on myeloid progenitor cells
- Pharmacological actions
  - Stimulate proliferation and differentiation
  - Activate phagocyte activity of mature neutrophils
  - Prolong neutrophils survival
  - Mobilize hematopoietic stem cells
  - Stimulate erythroid and megakaryocyte progenitors (GM-CSF)
- ADR
  - Flu-like symptom
  - Bone and muscle pain

Filgrastim: PK/PD Modeling



G-CSF	GM-CSF
Filgrastim, lenograstim, and pegfilgrastim	Molgramostim and sargramostim

- rhG-CSF is indicated for neutropenia associated with myelosuppressive cancer chemotherapy, bone marrow transplantation, and severe chronic neutropenia;
- rhG-CSF is also indicated to mobilize peripheral blood progenitor cells (PBPC) for PBPC transplantation; and
- rhG-CSF is indicated for the reversal of clinically significant neutropenia and subsequent maintenance or adequate neutrophil counts in patients with HIV infection during treatment with antiviral and/or other myelosuppressive medications.

- rhGM-CSF is indicated for neutropenia associated with bone marrow transplantation and antiviral therapy for AIDS-related cytomegalovirus.
- rhGM- CSF is also indicated for failed bone marrow transplantation or delayed engraftment and for use in mobilization and after transplantation of autologous PBPCs.
- rhGM-CSF used clinically to improve regeneration of these cells after bone marrow transplant

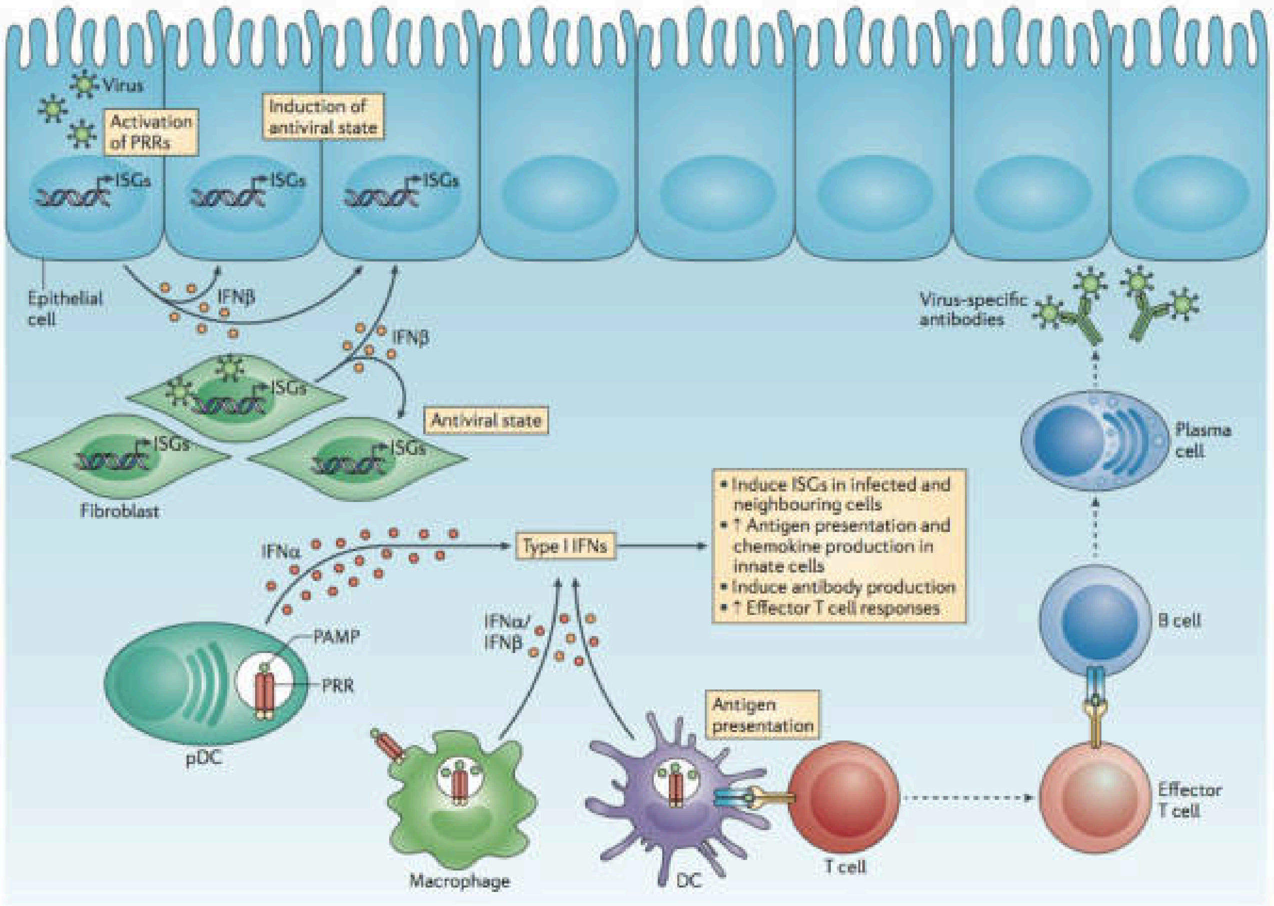
Other Hematopoetic Growth Factors

- Erythropoietin alpha (Epoetin alpha) (Procrit®)
  - Produced by recombinant DNA technology
  - Stimulates division and differentiation of erythroid progenitor cells
  - Used for anemia due to renal failure or cancer chemotherapy
  - Adverse effects include hypertension, headache, hypersensitivity reactions are rare
- Darbopoetin alpha (Aranesp®)
  - Recombinant long-acting erythropoietin (3X epoetin)

Interferon

- IFN – 2 types 3 families
  - Type 1 = IFN-α, IFN-β
    - Induced by viral infections
    - Leukocytes produce INF-α
    - Fibroblast and epithelial cells produce IFN-β
  - Type 2 = IFN-γ
    - T cell produce IFN-γ
- Activate JAK/STAT receptor superfamily on immune cells
- ADR:
  - Flu-like symptoms; Fatigues, Myalgia, Headache, Fever and chills
  - Depression
  - Myelosuppression

Interferon: Molecular of Action



Interferon stimulated gene (ISG)

Interferon Uses

Recombinant interferons	Company	First Indication	First approval
		Interferon-α	
IFN-α2a produced in *E. coli*; Roferon A®	Hoffmann–La Roche (Basel, Switzerland)	Hairy cell leukemia Chronic hepatitis B and C	1986 (EU and US)
IFN-α2b produced in *E. coli*; Intron® A; Viraferon®; Alfatronol®	Schering-Plough (Kenilworth NJ, USA)	Hairy cell leukemia Chronic hepatitis B and C	1986 (US and EU)
IFN-αcon1, synthetic type I IFN produced in *E. coli*; Infergen®	Amgen (Thousand Oaks, US), Yamanouchi Europe (Leiderdorp, The Netherlands, EU)	Chronic hepatitis B and C	2001 (US)
		Interferon-β	
IFN-β1a produced in CHO cells; Rebif®	Serono } (Geneva, Switzerland)	Relapsing/remitting multiple sclerosis	1998 (EU), 2002 (US)
IFN-β1a produced in CHO cells; Avonex®	Biogen (Cambridge, MA, USA)	Relapsing/remitting multiple sclerosis	1997 (EU), 1996 (US)
IFN-β1b, Cys17 Ser substitution; produced in *E. coli*; Betaferon®	Schering AG (Berlin, Germany)	Relapsing/remitting multiple sclerosis	1995 (EU)
IFN-β1b, Cys17 Ser substitution; produced in *E. coli*; Betaseron®	Berlex Labs/Chiron (Richmond/ Emeryville, CA, USA)	Relapsing/remitting multiple sclerosis	1993 (US)
		Interferon-γ	
Actimmune® (IFN-γ1b; produced in *E. coli*)	Genentech (San Francisco CA, USA), InterMune (Palo Alto, CA, USA)	Chronic granulomatous disease	1990 (US)

Adapted from Walsh G. Biopharmaceutical Benchmarks 2006. *Nature Biotechnology* (2006) 24:769–776

The NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

Adult-Onset Immunodeficiency in Thailand and Taiwan

Sarah K. Browne, M.D., Peter D. Burbelo, Ph.D., Ploenchan Chetchotisakd, M.D., Yupin Suputtamongkol, M.D., Sasisopin Kiertiburanakul, M.D., Pamela A. Shaw, Ph.D., Jennifer L. Kirk, B.A., Kamonwan Jutivorakool, M.D., Rifat Zaman, B.S., Li Ding, M.D., Amy P. Hsu, B.A., Smita Y. Patel, M.D., Kenneth N. Olivier, M.D., Viraphong Lulitanond, Ph.D., Piroon Mootsikapun, M.D., Siriluck Anunnatsiri, M.D., Nasikarn Angkasekwinai, M.D., Boonmee Sathapatayavongs, M.D., Po-Ren Hsueh, M.D., Chi-Chang Shieh, M.D., Ph.D., Margaret R. Brown, B.S., Wanna Thongnoppakhun, Ph.D., Reginald Claypool, R.N., Elizabeth P. Sampaio, M.D., Ph.D., Charin Thepthai, M.Sc., Duangdao Waywa, M.Sc., Camilla Dacombe, R.N., Yona Reizes, R.N., Adrian M. Zelazny, Ph.D., Paul Saleeb, M.D., Lindsey B. Rosen, B.S., Allen Mo, B.S., Michael Iadarola, Ph.D., and Steven M. Holland, M.D.

ABSTRACT

Disseminated NTM Disease

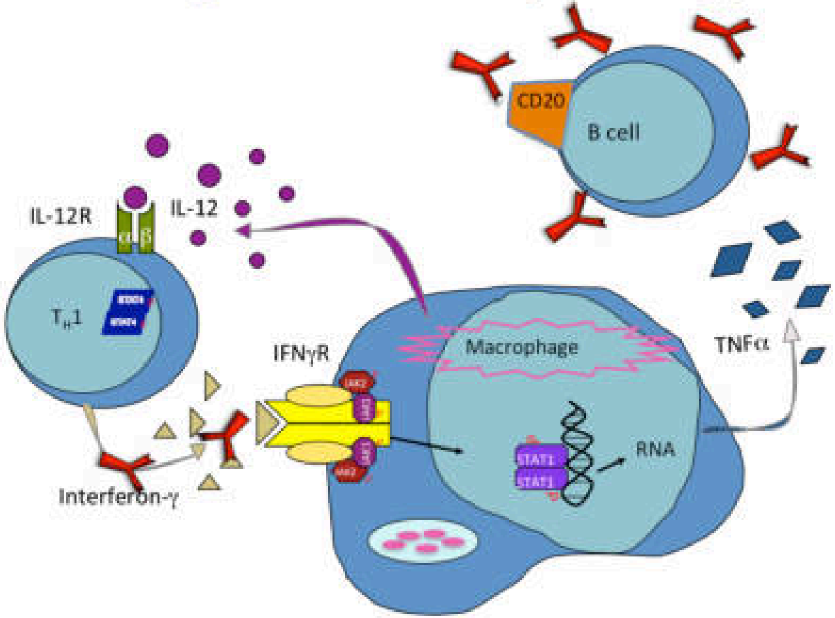

THE NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

Adult-Onset Immunodeficiency in Thailand and Taiwan

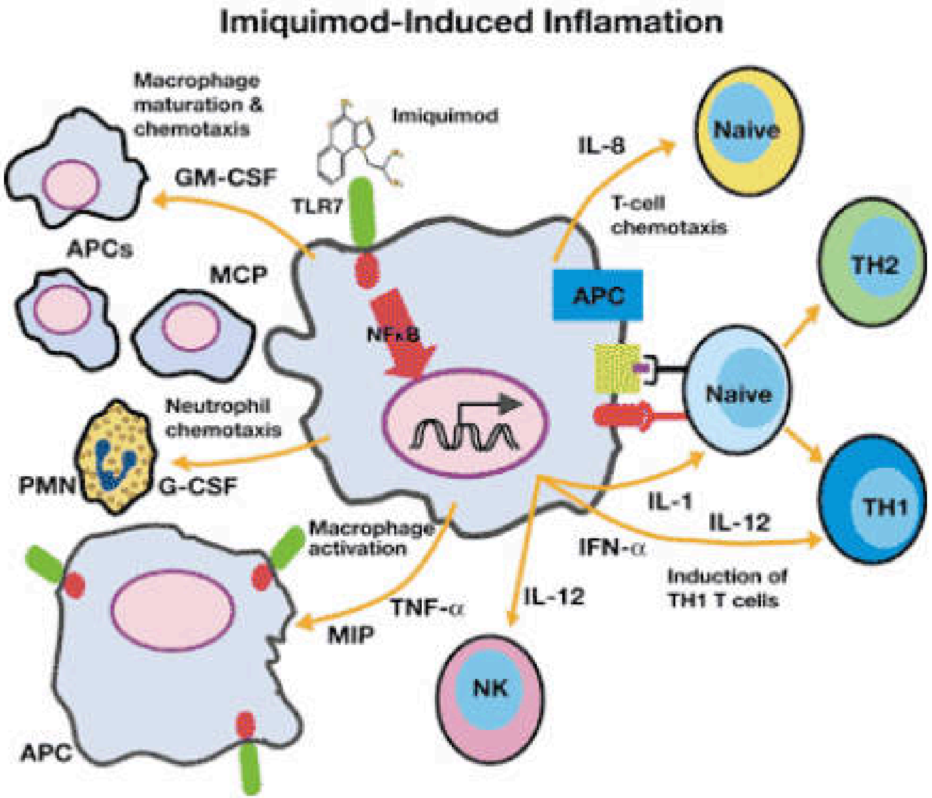
ABSTRACT

Insight of Immunopathology



Imiquimod

- Cream for local application
- Activate TLR-7 receptor of Langerhans cell and increase IFN-α release
- Indications – Wart and molluscum
- AE: Non-specific inflammation



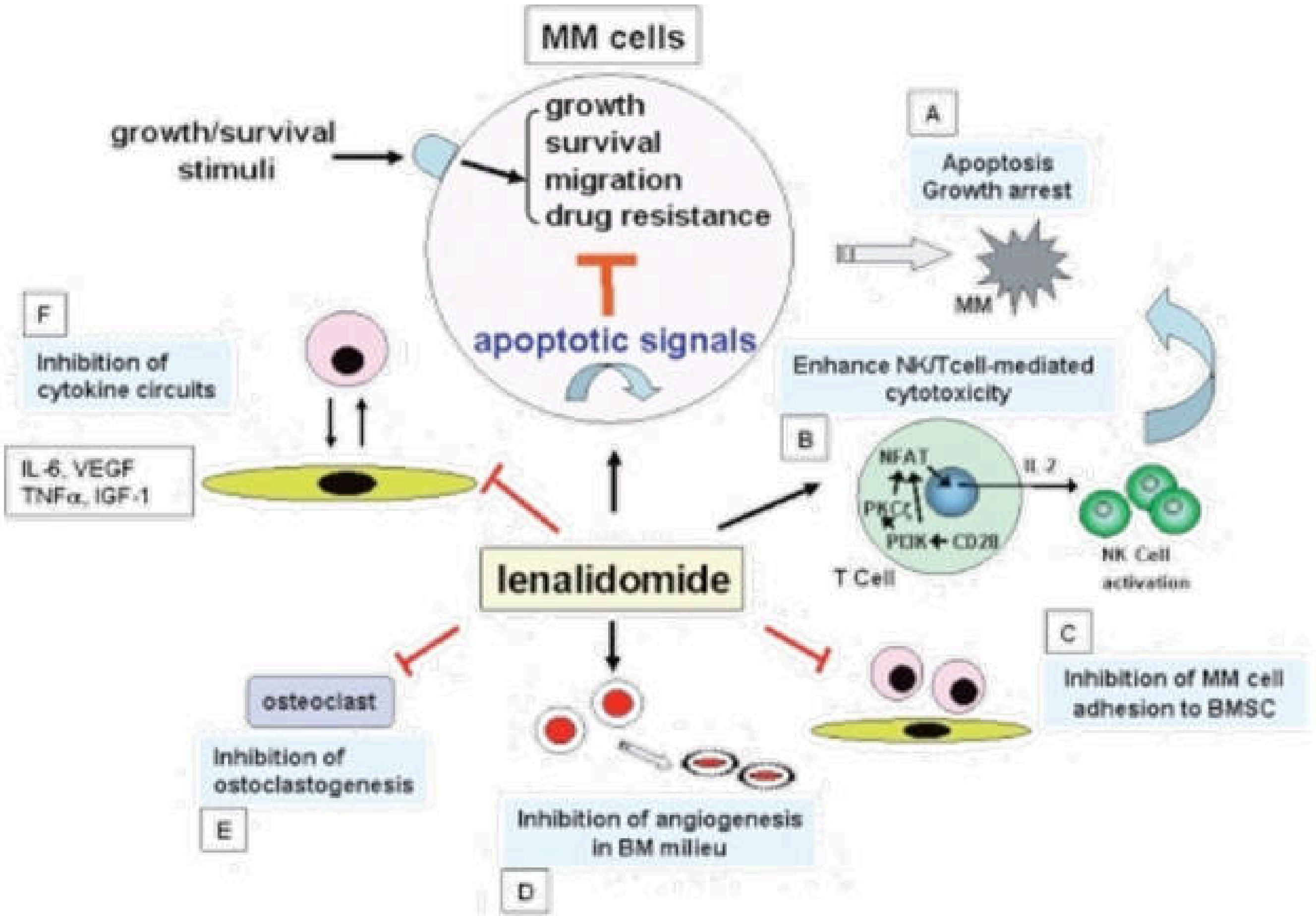
Thalidomide and Lenalidomide

- Thalidomide: Birth defect, Withdrawn
- Lenalidomide: Non-teratogenic derivative of Thalidomide
- Immunomodulatory effects
  - Binds cereblon and drives IKZF1/IKZF3 degradation, shutting down IRF4/c-Myc survival signaling in myeloma cells
  - Induces direct tumor cell-cycle arrest and apoptosis of multiple myeloma plasma cells
  - Boosts T- and NK-cell activation with Th1 cytokines, enhancing immune-mediated myeloma killing
  - Modulates the bone marrow milieu by inhibiting pro-inflammatory cytokines and angiogenesis

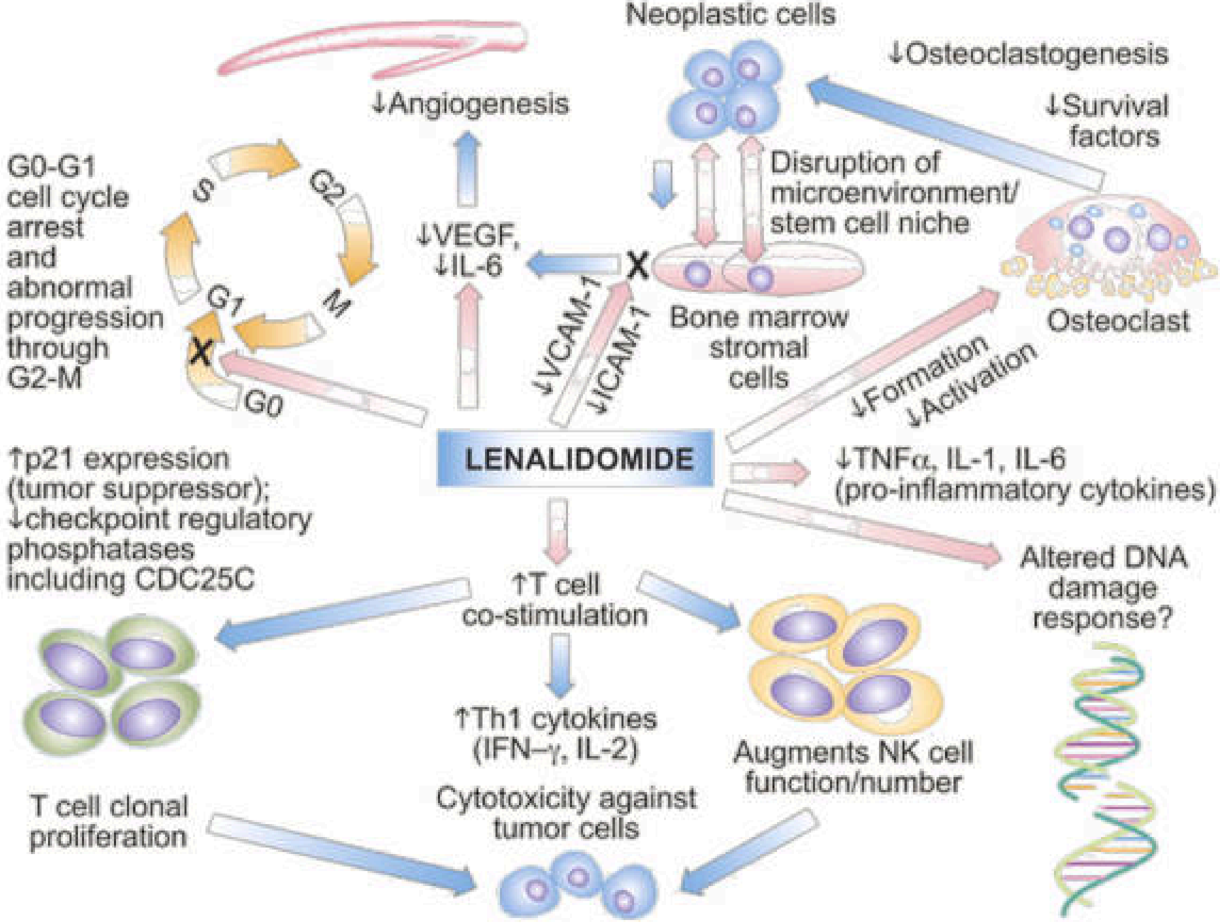
<u>USE:</u>

- Multiple myeloma (MM), AML

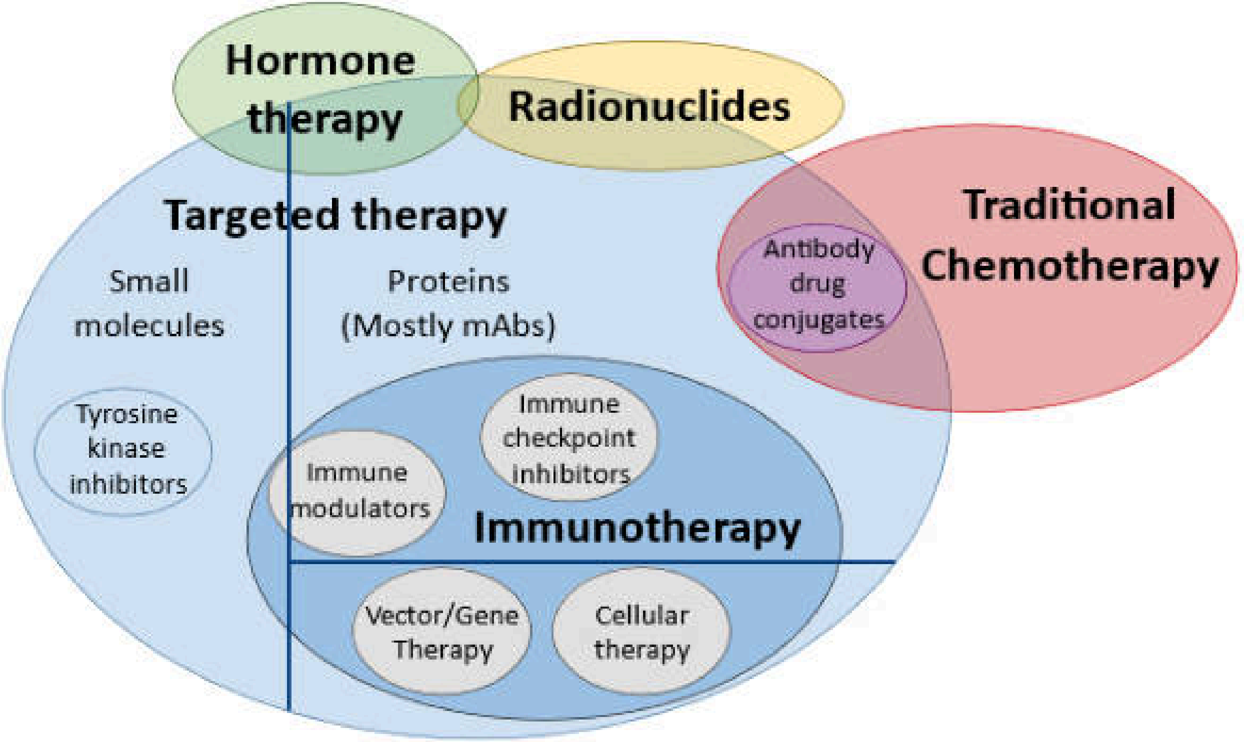
Lenalidomide: MOA in MM



Lenalidomide: MOA in AML

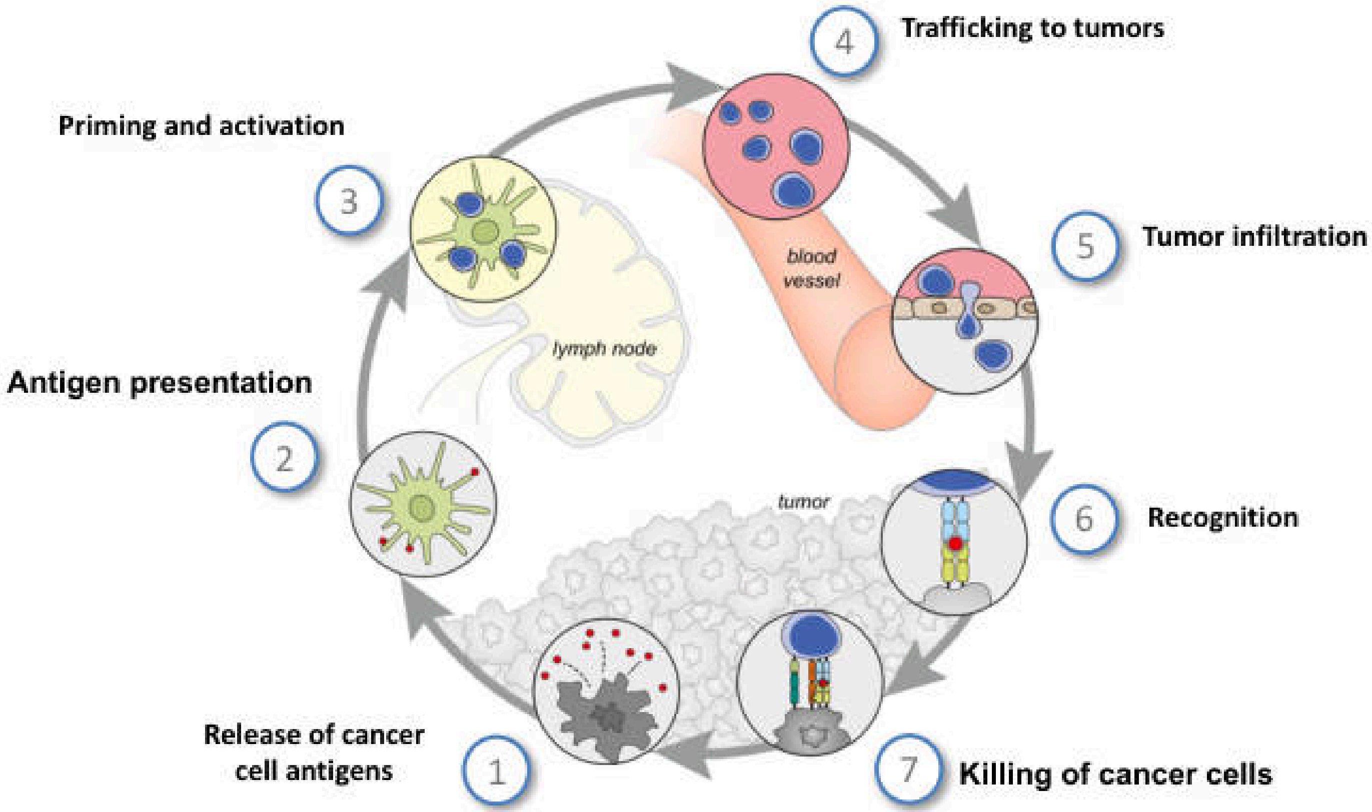


Classification of oncology medications, depicting overlapping classes



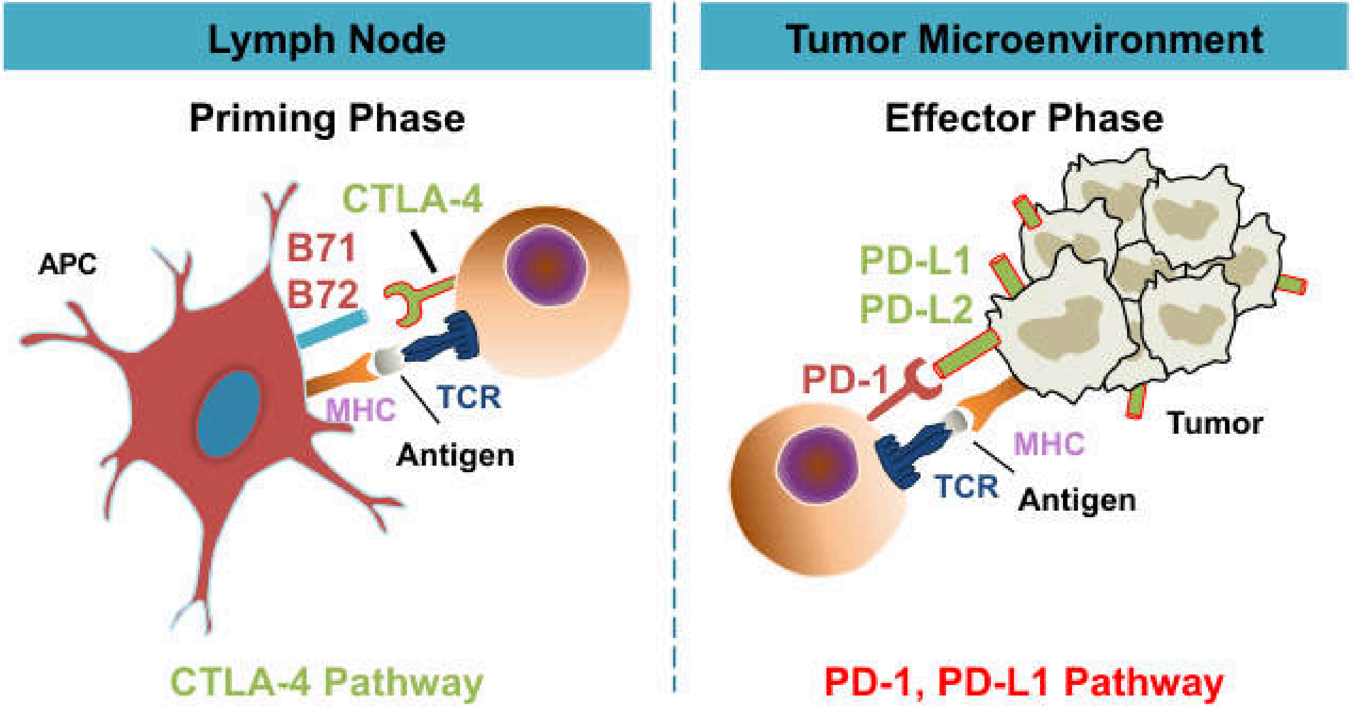
https://www.cancer.gov/publications/dictionaries/cancer-terms

The Cancer Immunity Cycle



Chen DS, et al. Oncology meets immunology: the cancer-immunity cycle. p. 1-10.

Focus on 2 Actionable Immune Synapses



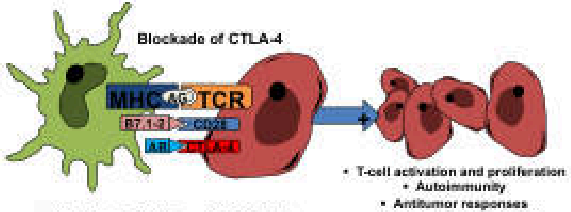
Ribas A. N Engl J Med. 2012;366:2517-2519.
Spranger S, et al. J Immunother Cancer. 2013;1:16.

irAEs: Considerations for Therapy

- Dose dependent
- Most irAEs develop during initial dosing period
  - Median time to onset: 5-9 wks
- Major categories of irAE associated with checkpoint inhibitors
  - Dermatologic
  - Gastrointestinal
  - Endocrine
  - Hepatic

- Graded by severity (CTCAE): 1 (mild) through 5 (death)
- Correlation between development of irAEs and treatment response
- Generally managed with use of corticosteroids or other immunosuppressive medications

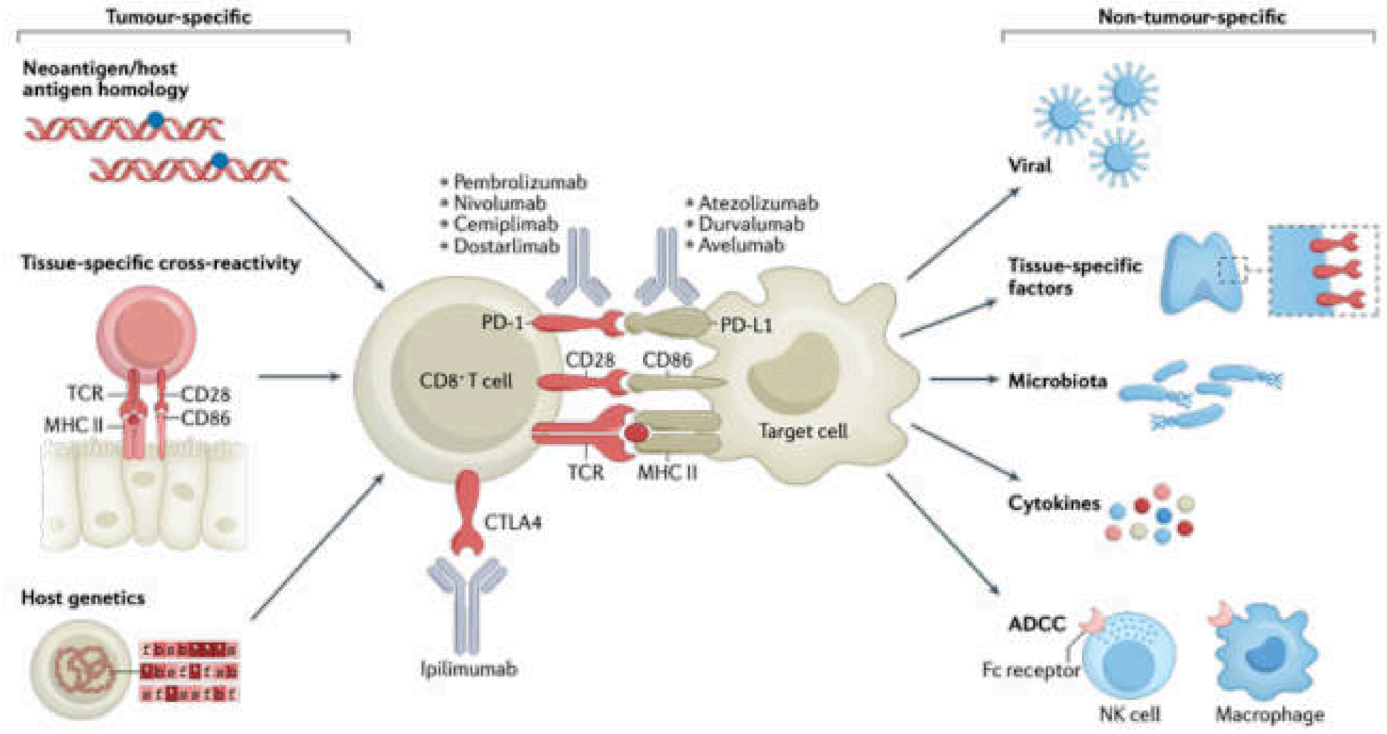
- "Achilles heel" of checkpoint inhibitors: autoimmunity via irAEs
- Unique toxicities of immunomodulators caused by dysregulation of the host immune system, similar to autoimmune disease



Dillard T, et al. Pituitary. 2010;13:29-38.

Michot JM, et al. 2016 Eur J Cancer. 54:139-148.

irAEs: Mechanism of Action



Nat Rev Clin Oncol **19**, 254–267 (2022)

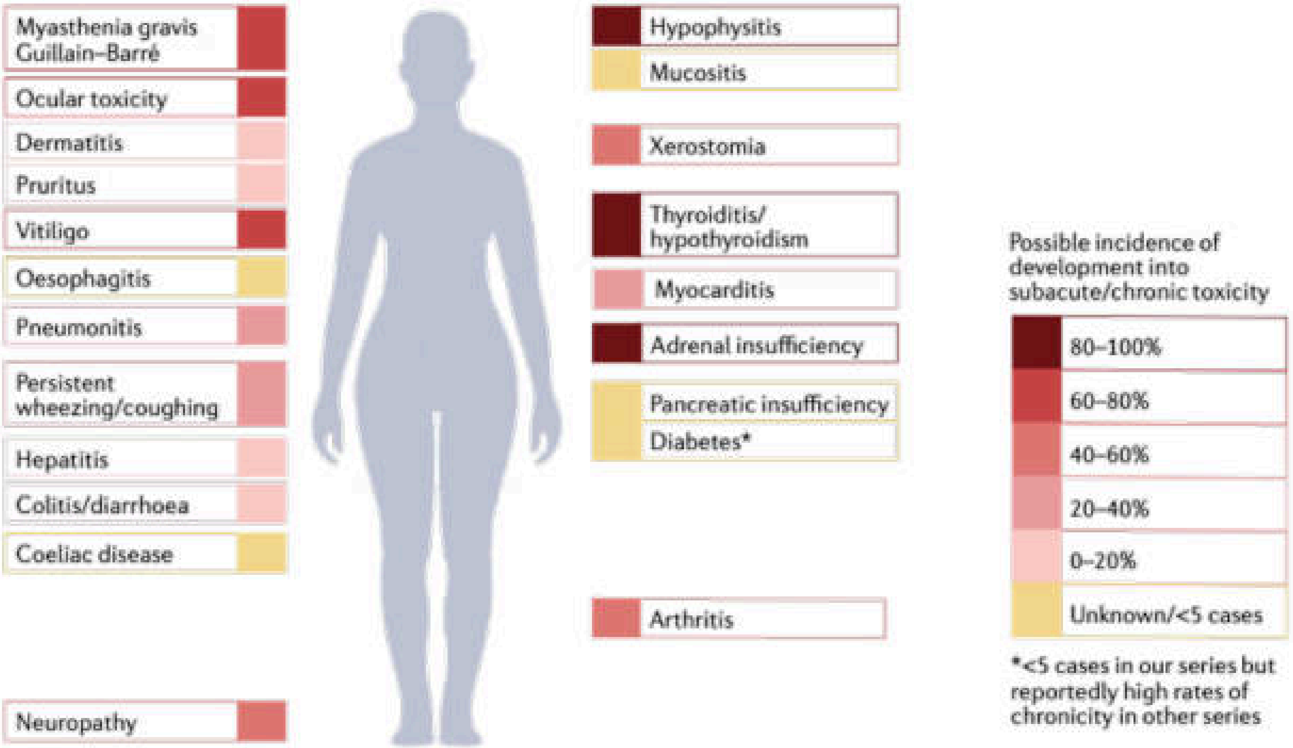
Serum auto-antibody assays with potential value for identifying IrAEs

Immune-related organ involved		Antibodies
Gastro-intestinal		None
Liver		Antinuclear antibodies (ANAs)
		Anti-smooth muscle, anti-liver kidney microsomal antibody type 1, anti-liver cytosol type 1
Lung		Antinuclear antibodies (ANAs)
		Rheumatoid factor
		Anti-centromere
		Extractable nuclear antigens (ENA): anti-Sm, anti-RNP; anti-Ro (SSA), anti-La (SSB); anti-Scl70, anti-Jo
Endocrine	Thyroid	Anti-thyroglobulin and anti-TPO
	Diabetes mellitus	Anti-GAD, anti-insulin, anti-carbonic anhydrase
	Addison's disease	Anti-21 hydroxylase
	Hypophysitis	Anti-pituitary
Skin		None
Polyarthritis		Antinuclear antibodies (ANAs)
		Anti-ENA: Anti-SSA, SSB, Sm
		Anti-CCP, complement fractions C3 C4 CH50
Renal		Antinuclear antibodies (ANAs)
		Complement fractions C3 C4 CH50
		Anti-neutrophil cytoplasmic (ANCA)
Haematologic syndromes		Antinuclear antibodies (ANAs)
		Coombs' erythrocyte test

IRAEs = immune-related adverse events; CCP = cyclic citrullinated peptide; GAD = Glutamate decarboxylase; RNP = ribonucleoprotein; Sm = Small nuclear ribonucleoprotein; SSA = Sjogren's syndrome-related antigen A; Scl = Sclerosis systemic; SSB = Sjogren's syndrome-related antigen B; TPO = Thyroid peroxidase.

J.M. Michot et al. / European Journal of Cancer 54 (2016) 139-148

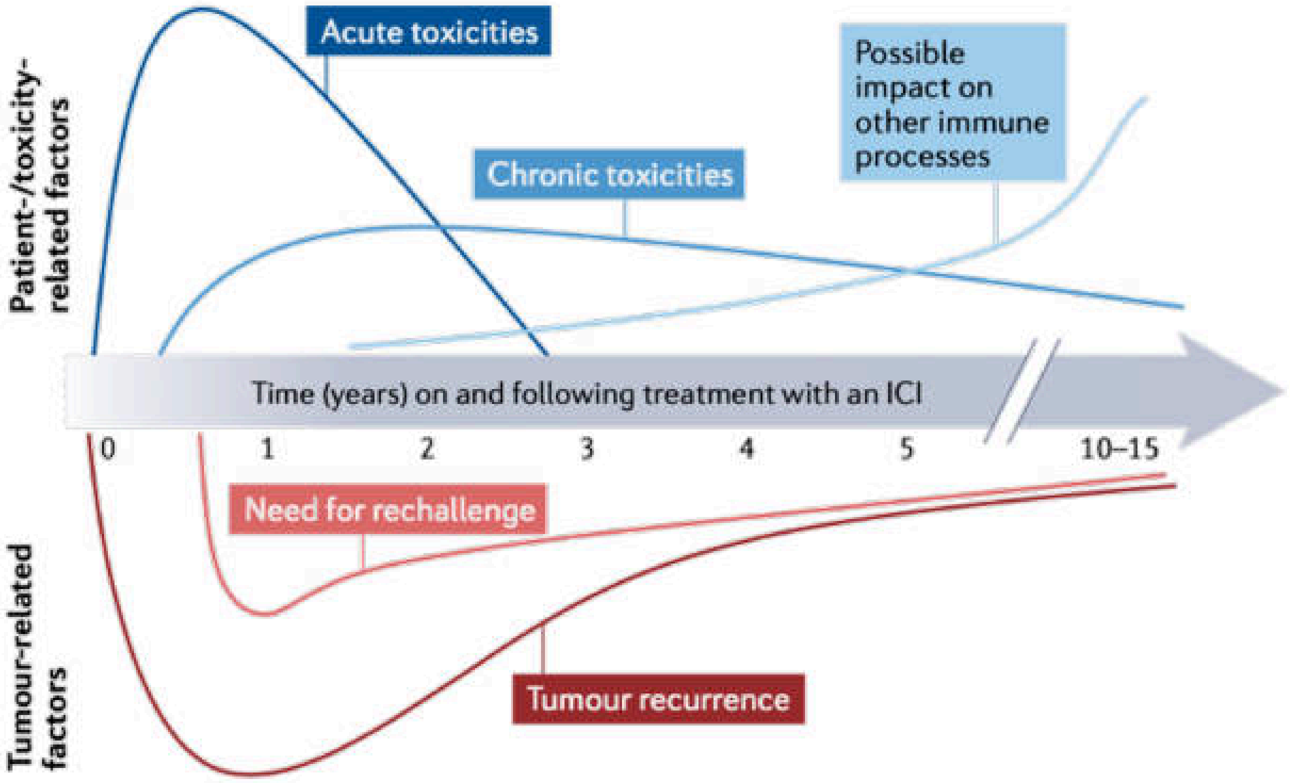
Chronic irAEs: ADRs



Possible frequencies of chronic immune-checkpoint inhibitor-induced toxicities

Nat Rev Clin Oncol **19**, 254–267 (2022)

Time course and potential importance of key issues throughout the course of treatment with immune-checkpoint inhibitors

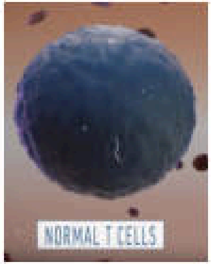


Nat Rev Clin Oncol **19**, 254–267 (2022)

Managing Immune-Related AEs

- Most immune-related AEs are reversible with immunosuppression through steroid treatment
  - Typically start with high-dose IV and then taper over 1-3 mos
  - Additional supportive care measures can be added for sustained immune-related AEs, such as infliximab for persistent diarrhea/colitis
  - Exception: adrenal insufficiency and hypothyroid need replacement hydrocortisone and levothyroxine, respectively, not immunosuppressive doses of steroids
- No evidence that intervening with steroids curtails antitumor efficacy of agent

Chimeric Antigen Receptor (CAR) T-Cell



CAR T-cells are modified T-cell by making new chimeric antigen receptor or artificial TCR from engineering.

CHIMERIC

The CAR protein is called “chimeric” because scientists engineer it using different components in order to get the T-cell to do what they want.

ANTIGEN

are proteins on the cancer cell that the T-cells are engineered to target.

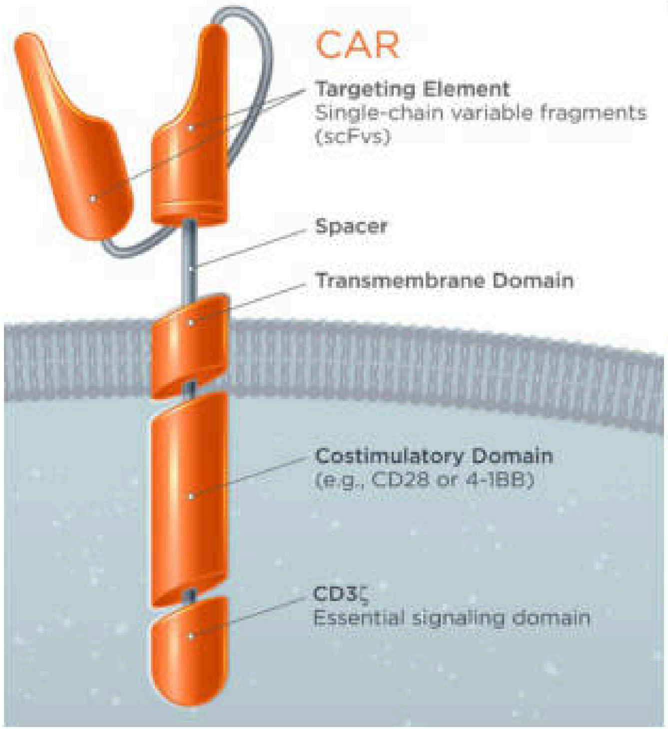
RECEPTOR

The CAR protein, which is added to the surface of the engineered T-cell, functions as a receptor.

T-cell

Re-engineered T-cells are this therapy’s primary weapons.

Chimeric Antigen Receptor (CAR) T-Cell



Extracellular domain

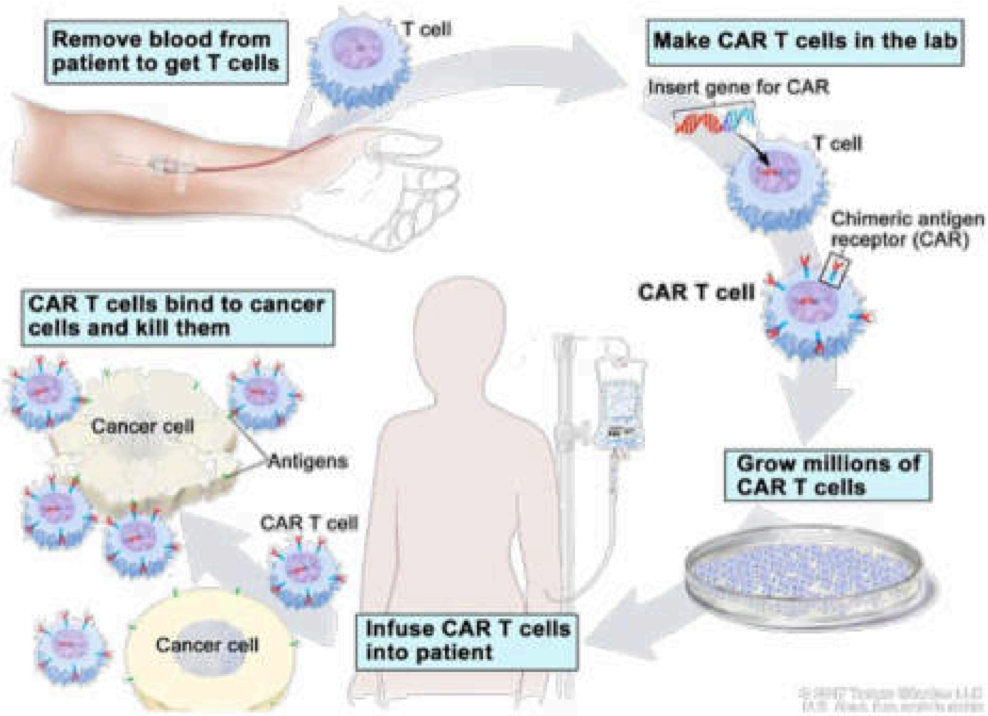
- Target element (scFv)
- Spacer

Trans-membrane domain

Intracellular signaling domain

- Co-stimulatory domain
- activating domain (CD3 zeta)

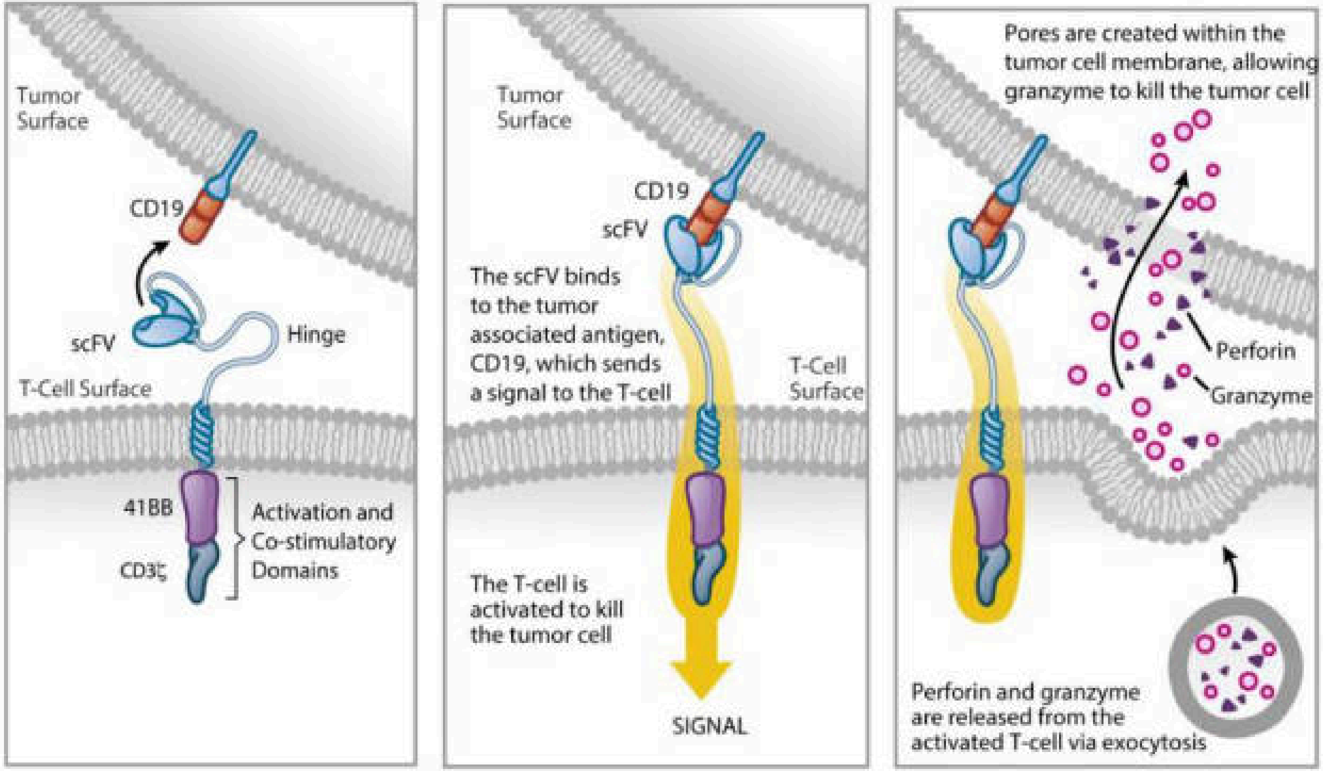
CAR T-cell Therapy



CAR T-cell therapy removes some of a person's T-cells and alters them to make them better able to fight cancer. The altered T-cells are then returned to the patient's body to go to work. Some researchers have called these re-engineered cells "a living drug"

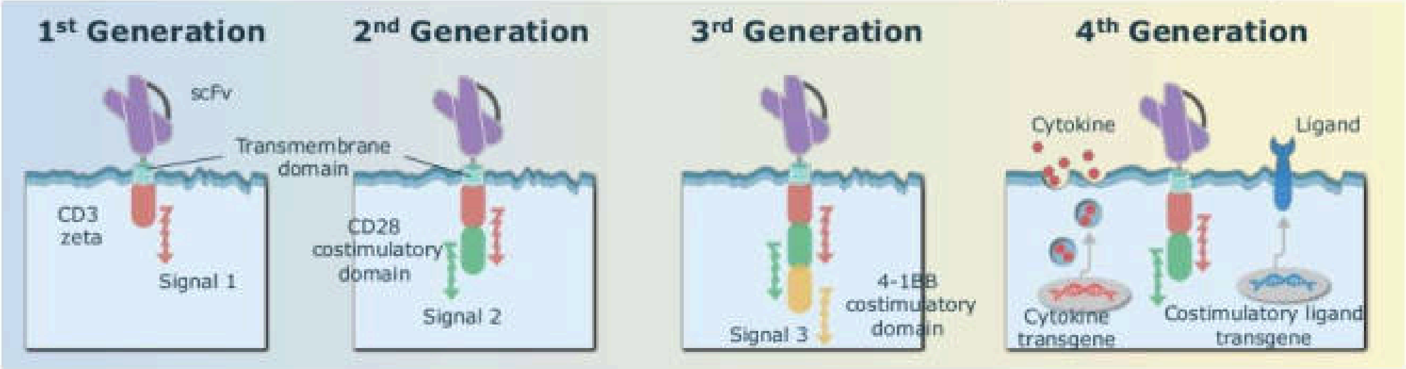
Chimeric Antigen Receptor (CAR) T-Cell

Mechanism of Action



Chimeric Antigen Receptor (CAR) T-Cell

Universal cytokine-mediated killing (TRUCKS)



	1st generation	2nd generation	3rd generation	4th generation
Add	-	1st generation + one co-stimulatory signal domain	1st generation + two co-stimulatory signal domain	2nd generation with addition of gene cassettes of cytokine or co-stimulatory ligands
Detail	Poor anti-tumor effect, short survival	Improving overall survival, proliferation, persistence	Enhance cytotoxicity and durability	Long term anti-tumor, high level of cytokine

Drugs

Chimeric Antigen Receptor (CAR) T-Cell

Advantages

- Tumor recognition independent of HLA (no HLA typing needed)
- Overcome some tumor immune escape issue such as MHC and endogenous antigen expression down regulation
- CAR can be engineered for target variety
- Highly effective against refractory ALL, CLL, DBCL
- "Living drug"
- Long term recognition
- 70-90% complete response rates

Disadvantages

- Hyperimmune activation :

  Cytokine Release Syndrome
- Cross reaction : Killing of normal B cell
- Resistance due to CD19 loss
- Not universal. Must be customizes for each individual
- High cost

Approved CAR T-cell Therapies

1. Tisagenlecleucel

FDA Approved Treatments,

2rd generation

-gen = -gene transfer of genetic material (transduced)

-lec = selection and enrichment manipulation

-leu = white blood cells

-cel = cellular therapy

Trade name: Kymriah™

Route of administration: Intravenous infusion

Indications: It is a CD19-directed genetically modified autologous T-cell immunotherapy with a lentivirus indicated for the treatment of patients up to 25 years of age with B-cell precursor acute lymphoblastic leukemia (ALL) that is refractory or in second relapse.

Approved CAR T-cell Therapies

1. Tisagenlecleucel

Dosage forms and strengths: Based on the patient's weight

- For patients 50 kg or less:

  0.2 to 5.0 x 10^6 CAR-positive viable T-cells per kg body weight

- For patients above 50 kg:

  0.1 to 2.5 x 10^8 CAR-positive viable T-cells

- The volume in an infusion bag ranges from 10 mL to 50 mL.
- Administered within 30 minutes of thawing at approx. 10-20 mL

Pharmacokinetics: T½ 16.8 days in responding patients

2.52 days in non responding patients

Approved CAR T-cell Therapies

1. Tisagenlecleucel

Preparing patient for KYMRIAH Infusion

Medication infusion

SUN	MON	TUE	WED	THU	FRI	SAT
31	1	2	3	4	5	6

Pre-treatment

lymphodepleting chemotherapy

Fludarabine

30 mg/m² IV

Cyclophosphamide

500 mg/m² IV

(for 2 days starting with the first dose of fludarabine)

Pre-medication

Do not use corticosteroids at any time except a lifethreatening emergency.

Acetaminophen

650 mg PO

Diphenhydramine

12.5 mg IV or PO

30-60 minutes before infusion

Administration

CAR-T-cell immunotherapy

Intravenous infusion at 10 mL to 20 mL per minute

Infuse 2 to 14 days after completion of pre-treatment

Approved CAR T-cell Therapies

2. Axicabtagene ciloleucel

FDA Approved Treatments, 2rd generation

Trade name: Yescarta™

Route of administration: Intravenous infusion

Indications:

A CD19-directed genetically modified autologous T-cell immunotherapy with retro virus for adult patients with relapsed large B-cell lymphoma after two or more other kinds of treatment, including diffuse large B-cell lymphoma (DLBCL).

Yescarta is **not indicated for** the treatment of patients with primary **central nervous system lymphoma.**

Approved CAR T-cell Therapies

2. Axicabtagene ciloleucel

Dosage forms and strengths :

- A single dose contains 2 $\times$ 10^6 CAR-positive viable T-cells per kg
- Maximum of 2 $\times$ 10^8 CAR-T-cells for patients 100 kg and above
- Approximately 68 mL suspension in an infusion bag

Pharmacokinetics :

Initial rapid expansion followed by a decline to near baseline levels by 3 months. Peak levels of anti-CD19 CAR T-cells occurred within the first 7-14 days after YESCARTA infusion.

Approved CAR T-cell Therapies

2. Axicabtagene ciloleucel

Preparing patient for YESCARTA Infusion

Medication infusion

SUN	MON	TUE	WED	THU	FRI	SAT
31	1	2	3	4	5	6

Pre-treatment

lymphodepleting chemotherapy

Fludarabine

30 mg/m² IV

Cyclophosphamide

500 mg/m² IV

5th, 4th, and 3rd day before infusion

Pre-medication

Avoid prophylactic use of corticosteroids

Acetaminophen

650 mg PO

Diphenhydramine

12.5 mg IV or PO

1 hour before infusion

Administration

CAR-T-cell immunotherapy

Infuse the entire contents of the YESCARTA bag within 30 minutes by either gravity or a peristaltic pump

Approved CAR T-cell Therapies

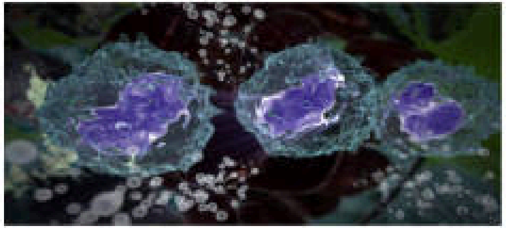
Possible Side Effects

1. Cytokine Release Syndrome (CRS)

produced multiple of cytokines

Symptoms: high fever, lower blood pressure, difficulty breathing, and may be associated with hepatic, renal, and cardiac dysfunction, and coagulopathy.

Do not administer drug to patients with active infection or inflammatory disorders.



Approved CAR T-cell Therapies

Possible Side Effects

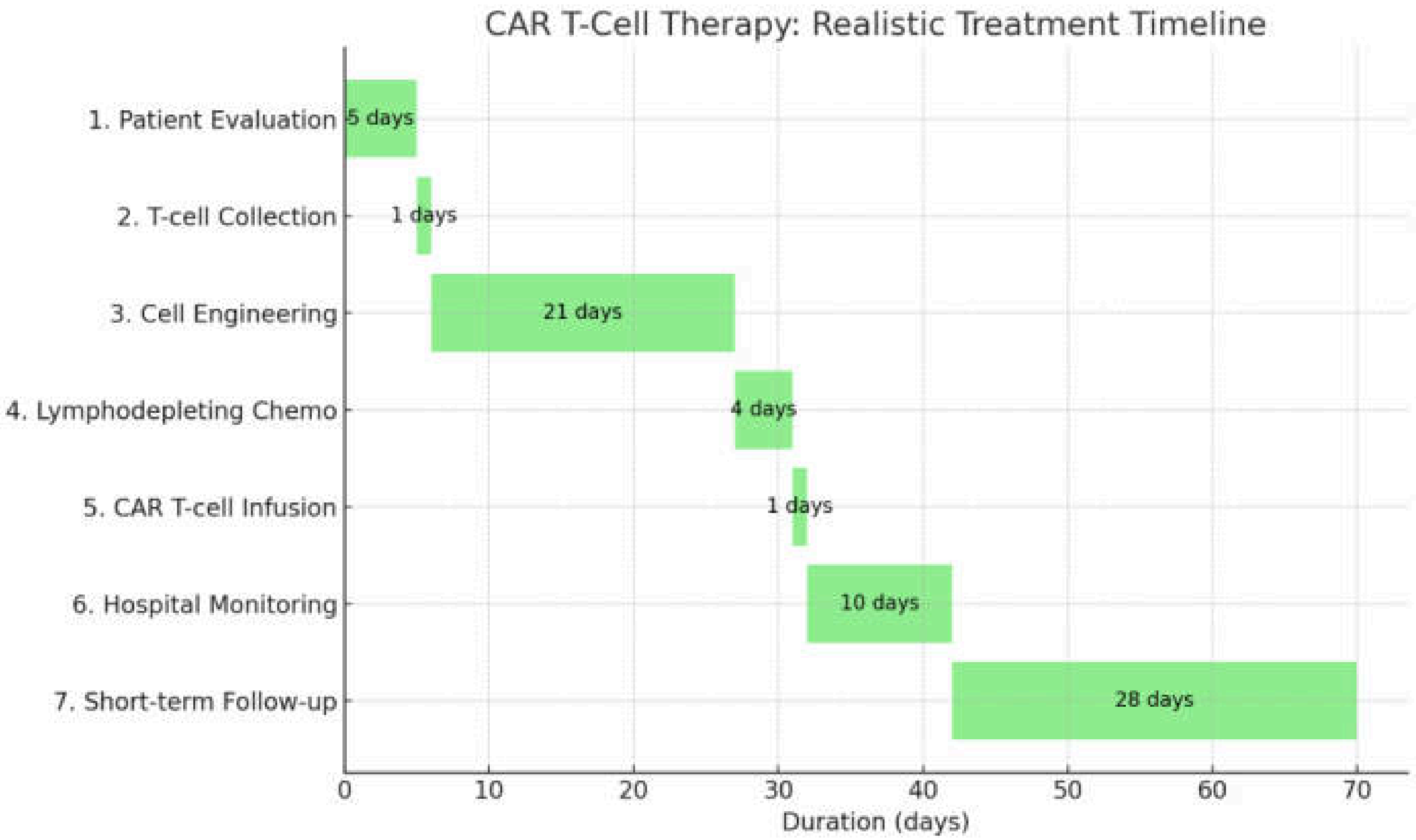
2. Neurological Toxicities

Symptoms: headache, encephalopathy, delirium, anxiety and tremor

3. Others

- Hypogammaglobulinemia
- infections-pathogen unspecified
- pyrexia (fever)
- decreased appetite
- headache
- hypotension
- nausea, diarrhea, vomiting

Approved CAR T-cell Therapies



Immunosuppressors

Corticosteroids

Target & MOA

Intracellular glucocorticoid receptor. Alters gene transcription. Decreases IL-1, IL-6, TNF, and IFN-gamma. Suppresses both innate and adaptive immunity.

Pharmacokinetics

Versatile dosing; utilizes both IV pulse and oral tapers. Rapid onset makes it ideal for acute control.

Major Use Cases

Induction therapy, rejection treatment, inflammatory flares, immune-related adverse event (irAE) treatment.

ADR & Monitoring Blueprint

Endocrine/Metabolic	Cardiovascular/Renal	Musculoskeletal	Psychiatric & Other
Hyperglycemia, weight gain	Hypertension	Osteoporosis	Mood changes, infection risk, GI/PUD, impaired wound healing
GLUCOSE, WEIGHT	BP	BONE HEALTH	PSYCH SYMPTOMS, INFECTION

Immunosuppressors

Pathway: Inhibit calcineurin -> Decrease NFAT activation -> Decrease IL-2 & T-cell activation.
Primary Use: Maintenance immunosuppression after transplant.

Cyclosporine	Tacrolimus
Target Cyclophilin-calcineurin complex.	**Target** FKBP-calcineurin complex.
PK Pearls Oral/IV. High drug-drug interactions. Trough-dependent monitoring in transplant practice.	**PK Pearls** More potent than cyclosporine. CYP metabolism (avoid grapefruit/pomegranate). Strict trough monitoring required.
Unique ADRs Hyperlipidemia, gout, hirsutism, **gingival hyperplasia**	**Unique ADRs** NODAT (New-Onset Diabetes After Transplant), **alopecia, hypomagnesemia, hypophosphatemia, diarrhea**
Shared ADRs Nephrotoxicity, HTN, hyperkalemia, neurotoxicity	**Shared ADRs** Nephrotoxicity, HTN, hyperkalemia, neurotoxicity.
Monitoring Labs TROUGH LEVELS, SCR/EGFR, K, MG, BP, NEURO EXAM	**Monitoring Labs** TROUGH LEVELS, RENAL FUNCTION, ELECTROLYTES, BP, GLUCOSE

Immunosuppressors

Purine Synthesis Inhibition

Agents: Azathioprine, Mycophenolate Mofetil (MPA)

Effect: Interferes with purine synthesis (Azathioprine) or inhibits IMPDH (MPA) to block de novo guanine synthesis. Decreases T/B-cell proliferation.

PK Pearls: Azathioprine interacts with allopurinol. MPA utilizes enterohepatic recycling and rarely requires trough monitoring.

ADR: Marrow suppression, GI upset, lymphoma risk, CMV. Azathioprine specific: Pancreatitis, hepatotoxicity.

Monitor: CBC, LFT

Pyrimidine Synthesis Inhibition

Agents: Leflunomide, Teriflunomide

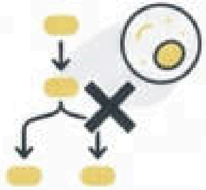
Target: Dihydroorotate dehydrogenase.

Pearls: Small molecules with long biologic effects. Used for autoimmune and inflammatory diseases.

ADR: Teratogenicity is a major class pearl. Hepatotoxicity, cytopenia.

Monitor: CBC, LFT, PREGNANCY TEST

mTOR Signaling Inhibition



Agents: Sirolimus, Everolimus

Target: Binds FKBP to inhibit mTOR downstream of the IL-2 receptor.

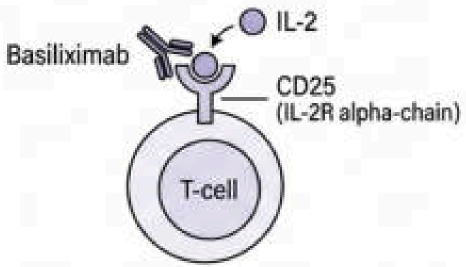
Pearls: CNI-sparing strategy. Delayed wound healing is a critical practical pearl.

ADR: Hyperlipidemia, thrombocytopenia, pneumonitis, proteinuria.

Monitor: LIPIDS, CBC, URINE PROTEIN, WOUND HEALING

Immunosuppressors

Basiliximab (Non-Depleting Receptor Blockade)



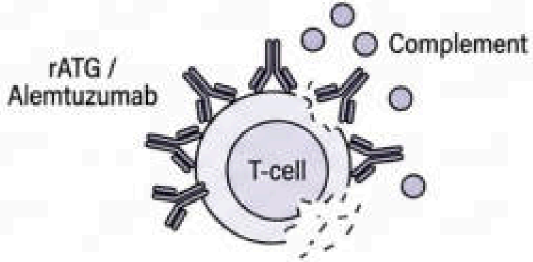
Target: CD25 (IL-2R alpha-chain) on activated T cells.

MOA: Competes with IL-2 at the receptor, blocking "Signal 3" without destroying the cell.

PK: Intermittent IV dosing; long duration relative to dose interval.

Clinical Profile: Used for induction in lower-risk transplants. Lower infection risk compared to depleting agents. Monitor for infusion reactions.

rATG / Alemtuzumab (Depleting Cellular Destruction)



Target: rATG hits multiple T-cell antigens; Alemtuzumab targets CD52.

MOA: Severe lymphocyte depletion and lysis via opsonization and complement-mediated mechanisms.

PK: Biologic effect lasts significantly longer than the dosing period. Utilized in high-risk settings.

Clinical Profile: High-risk induction; rejection treatment. High risk of infusion reactions, profound cytopenias, opportunistic infections. Alemtuzumab carries an autoimmunity risk.

Monitoring: CBC, PLATELETS, INFECTION SCREENING

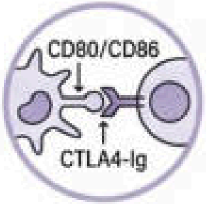
Immunosuppressors

Rituximab (B-Cell Depletion)

Target: CD20 on mature B cells.
MOA: Depletion via CDC, ADCC, and apoptosis.
Use: Desensitization, antibody-mediated rejection (AMR), B-cell malignancies.
Pearls & Risks: Long half-life monoclonal antibody. Repeat-course dosing concept. Risks include reactivation infections (Hepatitis B screening crucial), cytopenia, infusion reactions.

Bortezomib (Plasma Cell Apoptosis)

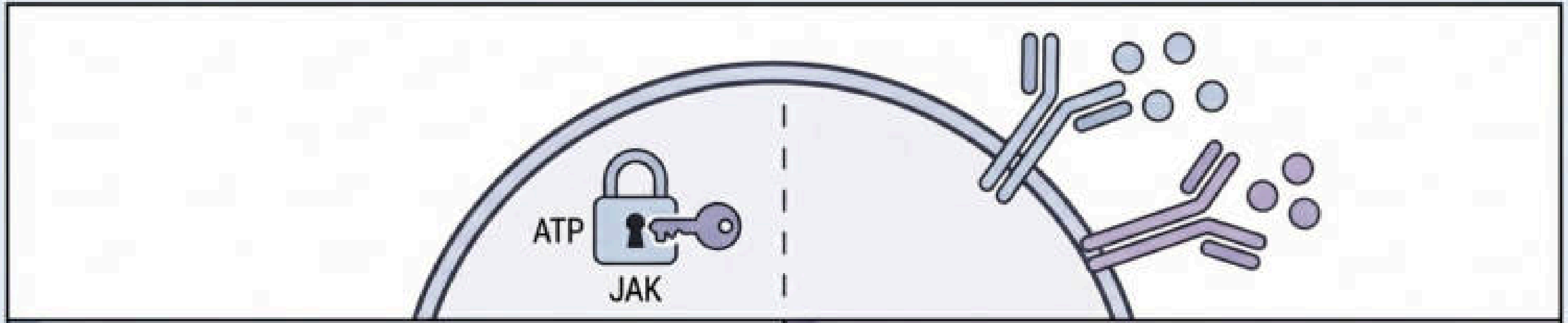
Target: 26S proteasome.
MOA: Induces ER stress and NF-kB depletion, reducing plasma cell survival.
Use: Antibody-mediated processes, desensitization. Short-course therapy.
Pearls & Risks: High risk of peripheral neuropathy, hypotension, GI toxicity, and thrombocytopenia.



Belatacept / Abatacept (Co-stimulation Blockade)

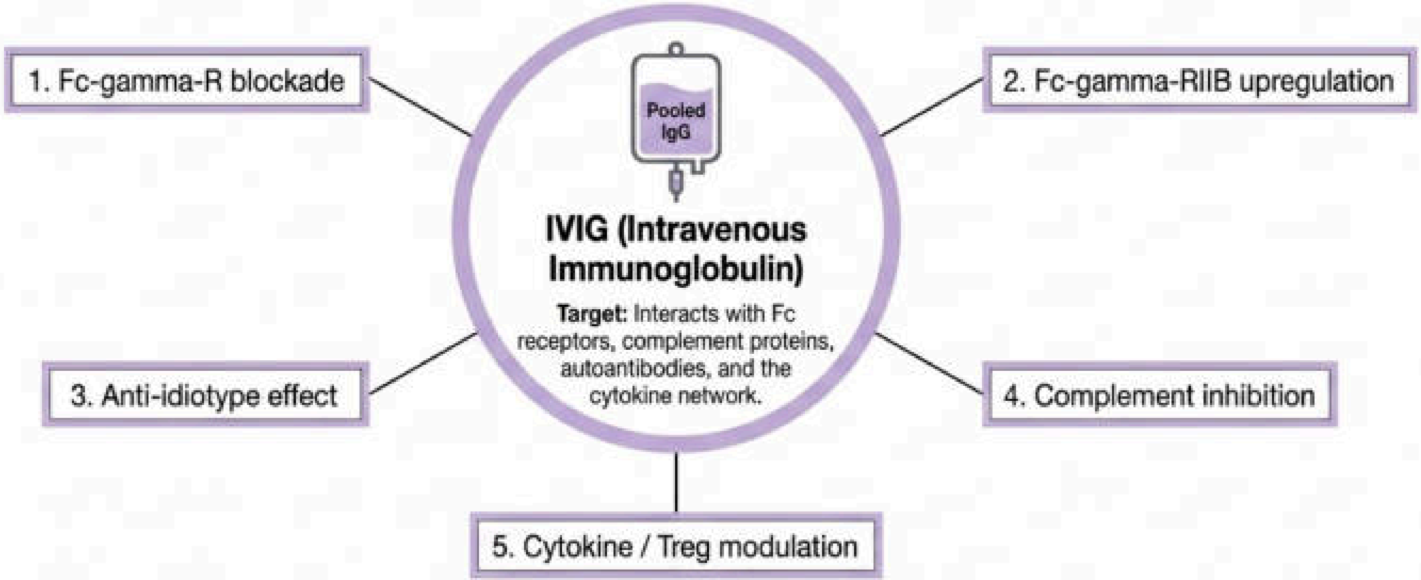
Target: CD80/CD86 on Antigen-Presenting Cells (APCs).
MOA: CTLA4-Ig fusion protein blocks 'Signal 2'.
Use: Belatacept for transplant maintenance; Abatacept for Rheumatoid Arthritis.
Pearls & Risks: Long half-life. Risks include Post-Transplant Lymphoproliferative Disorder (PTLD) — EBV screening required. Aphthous stomatitis, proteinuria.

Immunosuppressors



JAK Inhibitors (e.g., Tofacitinib)	**Anti-Cytokine / Targeted Biologics**
Target: Intracellular JAK-STAT signaling pathway. **MOA:** Competitive ATP-site inhibition, reducing signaling of multiple downstream cytokines. **PK:** Oral small molecule. Rapid elimination (t1/2 ~3 hours). Hepatic (70%) / Renal (30%) clearance. **Toxicity:** Herpes zoster, cytopenias, lipid elevation, GI perforation. **Monitor:** CBC, LFTS, LIPIDS, TB/VIRAL RISK	**Targets:** Extracellular TNF, IL-6R, IL-12/23, BLyS, IgE. **MOA:** Neutralize circulating cytokine ligands or block receptor signaling. **PK:** SC/IV administration. Long half-lives. Exhibit target-mediated or non-linear pharmacokinetics. **Toxicity:** Injection/infusion reactions, opportunistic infections, class-specific organ toxicity. **Use:** Widely deployed in RA, psoriasis, SLE, asthma, IBD.

Immunosuppressors



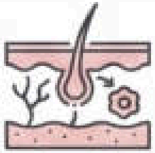
Clinical Applications: ITP, Kawasaki disease, primary immunodeficiency, transplant desensitization, passive immunity.

Critical PK Pearls: Formulations matter. Product properties (osmolality, sugar content, sodium, IgA content) directly influence clinical selection and tolerance.

Toxicity: Headache, aseptic meningitis, hemolysis, thrombotic events, osmotic nephropathy, bronchospasm.

Monitoring: RENAL FUNCTION, VOLUME STATUS, THROMBOSIS RISK

Immunostimulators



Systemic Adaptive (Vaccines)	
MOA	Active immunization to trigger adaptive immune response and memory (antigen-specific priming).
PK Pearls	Live vaccines induce stronger/longer immunity but carry strict precautions in immunocompromised hosts. Inactivated vaccines often require multiple doses/boosters.
ADR & Monitoring	Local/systemic reactogenicity. STRICT ADHERENCE to schedule, spacing, and contraindications is required.
Use Cases	Infectious disease prevention; some oncology applications.

Local Innate (Imiquimod)	
Target	TLR-7 on Langerhans cells and innate immune cells.
MOA	Stimulates innate immunity and drives local IFN-alpha production.
PK Pearls	Administered purely via topical/local routes.
ADR & Monitoring	Local inflammatory and skin reactions. MONITOR application site.
Use Cases	Warts, molluscum.

Immunostimulators

Filgrastim / GM-CSF (Myeloid Stimulation)

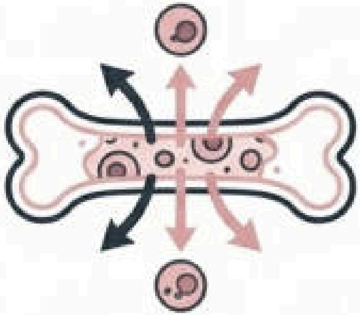
Target: Myeloid progenitor cytokine receptors (via JAK/STAT).

MOA: Stimulates proliferation, differentiation, neutrophil survival, and stem-cell mobilization.

PK: Exhibits both renal and target-mediated clearance.

Clinical Profile: Used for chemotherapy-induced neutropenia, chronic neutropenia, stem-cell mobilization.

ADR & Monitor: Bone and muscle pain, flu-like symptoms. Monitor CBC/ANC and clinical response.



Interferons (Systemic Activation)

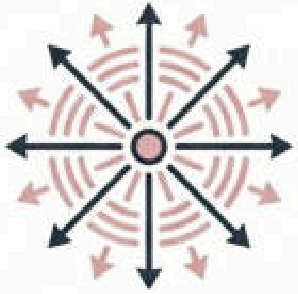
Target: IFN receptors on immune cells.

MOA: Activates JAK/STAT, inducing Interferon-Stimulated Genes (ISGs), enhancing antigen presentation and effector responses.

PK: Parenteral biologics offering sustained immunomodulatory effects.

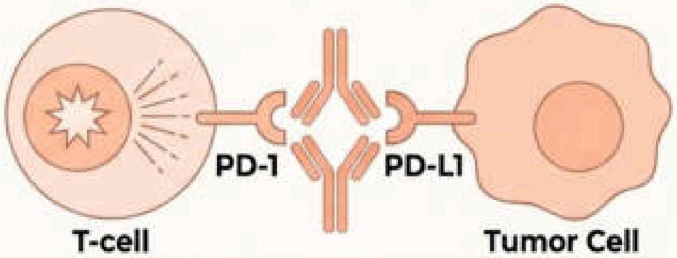
Clinical Profile: Used in viral diseases, multiple sclerosis, CGD, and some oncology.

ADR & Monitor: Flu-like syndrome, severe fatigue, depression, myelosuppression. Crucial to monitor MOOD/PSYCHIATRIC SYMPTOMS and CBC.



Immunostimulators

Checkpoint Inhibitors



Targets, MOA, PK Pearl

Targets: CTLA-4, PD-1, PD-L1.

MOA: Releases the immune 'brake,' unleashing and amplifying T-cell antitumor cytotoxic activity. Used broadly in cancer immunotherapy.

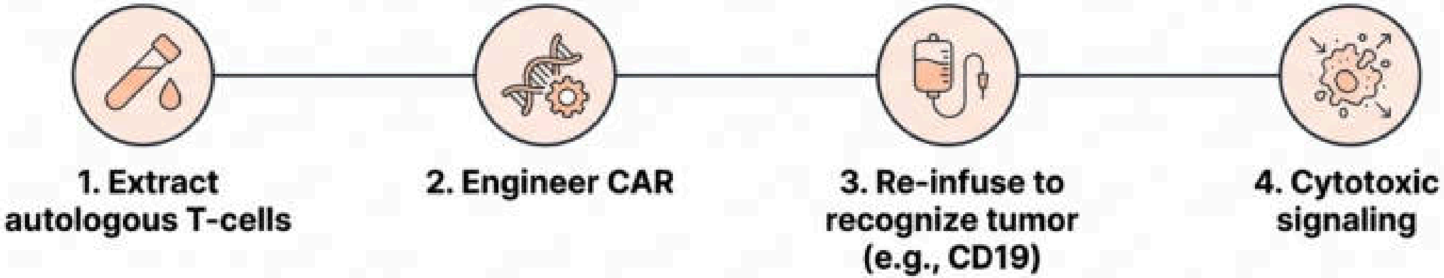
PK Pearl: Long half-life monoclonal antibodies. Crucially, toxicity may appear weeks to months after initiating or even stopping therapy.

⚠ Toxicity: Immune-Related Adverse Events (irAEs)

Can affect any organ system. Common manifestations include dermatitis, colitis, hepatitis, endocrinopathy, and pneumonitis.

Monitoring Blueprint:
Systematic symptom review
LFTs, THYROID FUNCTION TESTS (TFTs), GLUCOSE
Ongoing gastrointestinal, pulmonary, and endocrine surveillance.

Immunostimulators



Therapy: CAR-T Cell Therapy

MOA: Engineering of autologous T cells to express a Chimeric Antigen Receptor (CAR) that recognizes specific tumor antigens, killing tumor cells via cytotoxic signaling. Used for refractory B-cell malignancies.

The "Living Drug": Expands in vivo. Half-life and persistence vary significantly depending on the patient's clinical response.

Acute Toxicity & Protocol

CRS: Cytokine Release Syndrome. Massive systemic inflammatory response.

ICANS: Immune effector cell-associated neurotoxicity syndrome.

Other: Hypogammaglobulinemia, severe infections.

Monitoring: Requires strict inpatient observation post-infusion, rigorous CRS/ICANS monitoring, CBC, and infection surveillance.

Toxicity Crosswalk: Recognizing Adverse Event Signatures

	Nephrotoxicity / Renal Impairment	**Cyclosporine, Tacrolimus**, IVIG (osmotic), **Sirolimus** (proteinuria).
	Bone Marrow Suppression / Cytopenias	**Azathioprine, Mycophenolate** (MPA), **Leflunomide**, rATG/Alemtuzumab, **Rituximab**, **JAK inhibitors.**
	Metabolic / Endocrine Disruption	**Corticosteroids** (Hyperglycemia), **Tacrolimus** (NODAT), **Cyclosporine/Sirolimus** (Hyperlipidemia), **Checkpoint Inhibitors** (irAE endocrinopathy).
	Neurotoxicity	**Cyclosporine, Tacrolimus, Bortezomib** (Peripheral), CAR-T (ICANS).
	Infusion / Administration Reactions	**Basiliximab**, rATG/Alemtuzumab, Rituximab, Belatacept, IVIG, Anti-cytokines.

Clinical Indication Blueprint

Solid Organ Transplant (Induction & Maintenance)

- **Induction: rATG, Alemtuzumab** (high risk), **Basiliximab** (low risk), Corticosteroids.
- **Maintenance: Tacrolimus, Cyclosporine,** Mycophenolate (MPA), Azathioprine, Sirolimus, Belatacept.

Autoimmune & Inflammatory Diseases

- **Systemic/Flare: Corticosteroids, IVIG.**
- **Targeted Chronic: JAK Inhibitors, Anti-Cytokines** (TNF, IL-6), **Leflunomide, Abatacept.**

Oncology & Cellular Therapy

- **Solid Tumors:** Checkpoint Inhibitors (PD-1, CTLA-4).
- **Refractory B-Cell:** CAR-T, Rituximab, Bortezomib.

Specialty Interventions (Desensitization & Hematology)

- **Desensitization:** IVIG, Rituximab, Bortezomib.
- **Neutropenia Recovery:** Filgrastim / GM-CSF.